Numerical evaluation, matrix representation, and cascade application of decay feed-down

Notes accompanying `DecaySpectra.jl` and `CascadeTables.jl`

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These notes accompany `DecaySpectra.jl` and `CascadeTables.jl`. For the decays with two, three, or four daughters in the main physics notes (whose equation numbers are used below), they document how each map is (i) reduced to a form that fixed-order Gauss–Legendre quadrature evaluates rapidly, (ii) precomputed as a matrix acting on a parent spectrum sampled on a fixed grid, and (iii) composed through the full resonance cascade.

How to read this. The presentation is *topic by topic*, and within each topic the *analytic* structure is kept separate from its *numerical* treatment: first what an integral *is* — its domain, its singularities, any closed-form structure — then how it is *computed*. The topics build on one another:

- the two-body kernel and its quadrature (Sec. 2);
- the effective-mass average that carries three- and four-body decays (Sec. 3);
- finite resonance widths (Sec. 4);
- the matrix representation for boost-invariant sources (Sec. 5);
- full-list table construction, cascade ordering, memory, and persistence (Sec. 6);
- its block-Toeplitz generalization to rapidity-dependent sources (Sec. 7);
- the validated numerical domain, regression evidence, and known boundary cases (Sec. 8).

One device recurs throughout and is worth keeping in mind: every multi-body or finite-width complication is an *average of the same two-body-kernel implementation* over an effective-mass or spectral distribution. A multi-body decay changes only the cheap averaging weight at fixed parent mass; a finite parent width evaluates that kernel at several parent-mass nodes and folds the results into one same-shape matrix.

1 The decay map is linear

After the analytic azimuthal integration, the daughter spectrum of the two-body decay $a \to b+c$ is [Eq. (32)]

$$\mathcal{F}_b(p_T,\eta_p,\phi_p) = \frac{B\,m_a}{4\pi\,p^a_{b|c}} \int_0^\infty q_T\,\mathrm{d}q_T \int_{-\infty}^\infty \mathrm{d}\eta_q\, \frac{\Theta(Q^2-D^2)}{\sqrt{Q^2-D^2}} \Big[\mathcal{F}_a(q_T,\eta_q,\phi_p+\alpha)+\mathcal{F}_a(q_T,\eta_q,\phi_p-\alpha)\Big], \tag{1}$$

with $M = m_{aT} m_{bT}$, $Q = q_T p_T$, $C = m_a E^a_{b|c}$, $D = M\cosh(\eta_q - \eta_p) - C$, $Z = D/Q$, $\alpha = \arccos Z$. The three-body decay [Eq. (59)] is the same kernel averaged over the effective mass μ of the $c + d$ pair with weight $p^a_b(\mu)\, p^{\tilde{c}}_c(\mu)/\mathcal{N}_{a\to bcd}$ [Eqs. (45)–(48)]. In Fourier modes, $\mathcal{F} = \sum_n \mathcal{F}_n e^{in\phi}$ [Eq. (62)], the two azimuthal roots produce the Chebyshev factor $2T_n(Z)$ [Eqs. (40), (41), (61)].

The essential observation is that (1) is *linear* in $\mathcal{F}_a$. Once the parent spectrum is represented by its values on a fixed grid, the map is a matrix, built once per decay and per mode; applying it to any source is a matrix–vector product.

2 The two-body kernel

Both decays reduce to the double integral

$$K(E_b) = \int q_T \,\mathrm{d}q_T \int \mathrm{d}\eta_q \,\frac{\Theta\big(Q^2 - D^2\big)}{\sqrt{Q^2 - D^2}}\,\big[\text{bracket}\big], \qquad p_b = \sqrt{E_b^2 - m_b^2}, \quad C = m_a E_b, \tag{2}$$

where the bracket is the source sum of (1) (or $2T_n(Z)\mathcal{F}_{a,n}$ in mode form). The integrand lives on a compact region with inverse-square-root boundary singularities. This section has two halves. The analytic half (Sec. 2.1) fixes that region in closed form and locates every non-smooth feature: the $1/\sqrt{\ }$ window edges and a single logarithmic singularity where the domain changes topology. The numerical half (Sec. 2.2) removes each by one substitution, so that plain tensor-product Gauss–Legendre quadrature converges rapidly, and reports the measured accuracy.

2.1 Analytic structure: domain and singularities

2.1.1 The rapidity-difference window

What the window is. Before the algebra, the physical content. The analytic azimuthal integration of the main notes rests on the on-shell δ-function $\delta(q\cdot p + m_a E_b)$ [Eq. (4)], which at fixed rapidities fixes the relative azimuth through $\cos\Delta\phi = Z$ [Eq. (19)], with

$$Z = \frac{M\cosh\Delta\eta - C}{Q} = \frac{D}{Q}. \tag{3}$$

A real opening angle $\Delta\phi$ exists *iff* $|Z| \le 1$ [Eq. (21)] — and $\Theta(Q^2 - D^2) = \Theta(1 - Z^2)$ is exactly the indicator of that condition. So the "$\Delta\eta$ window" answers a kinematic question: *for a daughter observed at (p_T, η_p), which parent momenta (q_T, η_q) can produce it?* The parent must sit at a rapidity separation $\Delta\eta$ for which some azimuthal orientation places the daughter at p_T; too small or too large a $\cosh\Delta\eta$ makes $|Z| > 1$ and no real $\Delta\phi$ exists. The endpoints $Z = \pm 1$ are the grazing configurations $\Delta\phi = 0, \pi$ (daughter emitted parallel or antiparallel to the parent's transverse momentum), and the $1/\sqrt{Q^2 - D^2} = 1/(Q\sqrt{1 - Z^2})$ factor is the Jacobian $|\mathrm{d}\cos\Delta\phi/\mathrm{d}\Delta\phi|^{-1}$ that diverges integrably there, as at any turning point.

The bounds. Throughout, $M = m_{aT} m_{bT} > 0$, $Q = q_T p_T \ge 0$ (with $Q > 0$ for interior nodes, $q_T, p_T > 0$), and $C = m_a E_b > 0$; all three depend on q_T and/or p_T but not on $\Delta\eta$. At fixed q_T the integrand is supported where the argument of the Θ function is positive:

$$Q^2 - D^2 \ge 0 \iff |D| \le Q \iff -Q \le M\cosh\Delta\eta - C \le Q. \tag{4}$$

Adding C and dividing by $M > 0$ (both operations preserve the inequalities) gives the two-sided condition

$$\frac{C-Q}{M} \;\le\; \cosh\Delta\eta \;\le\; \frac{C+Q}{M}. \tag{5}$$

This must be intersected with the range of the hyperbolic cosine, $\cosh\Delta\eta \ge 1$. Two of the three bounds compare trivially: $(C-Q)/M \le (C+Q)/M$ always (they differ by $2Q/M \ge 0$). Therefore

$$\cosh\Delta\eta \;\in\; \Big[\max\Big(1, \tfrac{C-Q}{M}\Big),\; \tfrac{C+Q}{M}\Big], \tag{6}$$

and this interval is nonempty *iff* its upper end clears the lower: since $\max(1,(C-Q)/M) \le (C+Q)/M$ reduces — whichever branch the max takes — to the single condition

$$\frac{C+Q}{M} \;\ge\; 1 \;\iff\; M \;\le\; C+Q, \tag{7}$$

because the branch $(C-Q)/M \le (C+Q)/M$ holds unconditionally. Condition (7) is the origin of the q_T band of Sec. 2.1.2.

To translate (6) into a set in $\Delta\eta$: on $\Delta\eta \ge 0$ the map $\Delta\eta \mapsto \cosh\Delta\eta$ is strictly increasing and continuous, so the preimage of the interval (6) is the interval $[\eta_{\rm lo}, \eta_{\rm hi}]$ with

$$\eta_{\rm hi} = \operatorname{arccosh}\frac{C+Q}{M}, \qquad \eta_{\rm lo} = \begin{cases} \operatorname{arccosh}\dfrac{C-Q}{M}, & C-Q > M, \\ 0, & C-Q \le M, \end{cases} \tag{8}$$

(the second case includes $C-Q \le 0$, where the lower constraint in (5) is vacuous since $\cosh \ge 1 > (C-Q)/M$). Since cosh is even, the full solution set is symmetric,

$$\big\{\Delta\eta :\; |\Delta\eta| \in [\eta_{\rm lo}, \eta_{\rm hi}]\big\} = \big[-\eta_{\rm hi}, -\eta_{\rm lo}\big] \cup \big[\eta_{\rm lo}, \eta_{\rm hi}\big] : \tag{9}$$

two disjoint intervals when $\eta_{\rm lo} > 0$ (the parent *must* be at different rapidity than the daughter), merging into the single interval $[-\eta_{\rm hi}, \eta_{\rm hi}]$ when $\eta_{\rm lo} = 0$. The change between the two topologies as q_T varies is the origin of the logarithmic singularity of Sec. 2.1.3.

Geometrically the two regimes read as follows. When $\eta_{\rm lo} > 0$ (i.e. $C-Q > M$, meaning C is large: the parent is slow or the daughter hard) the parent cannot sit at the same rapidity as the daughter — a co-moving parent would have to emit the daughter with $|Z| > 1$ — so the allowed set is two rapidity bands straddling η_p, symmetric about it. As q_T increases, Q grows, $C-Q$ eventually drops below M, the inner gap closes ($\eta_{\rm lo} \to 0$), and the bands merge: now a parent *at* the daughter's rapidity can feed it (with $\Delta\phi$ given by the interior value $Z(0) = (M-C)/Q \in (-1,1)$). The merging point is exactly q_T^* of Sec. 2.1.3. The upper edge $\eta_{\rm hi}$ always exists (some maximal rapidity separation beyond which the boost carries the daughter past p_T for every azimuth) and is always a genuine $1/\sqrt{\ }$ singularity; the lower edge is singular only in the two-band regime. Both rapidity signs $\Delta\eta \gtrless 0$ contribute; they are evaluated separately because the source need not be symmetric in rapidity. At the window edges $Z = \pm 1$ and $Q^2 - D^2$ vanishes *linearly* in $\Delta\eta$: writing $Q^2 - D^2 = (Q-D)(Q+D)$ (the factored form is also what is computed — it avoids the cancellation of subtracting two large squares), one factor vanishes at each edge while $\mathrm{d}D/\mathrm{d}\Delta\eta = M\sinh\Delta\eta \ne 0$ there, so the zero is simple and the integrand diverges as an integrable $1/\sqrt{\ }$. When $\eta_{\rm lo} = 0$ (merged window) the lower edge is *not* singular — $Z(0) = (M-C)/Q$ lies inside $(-1,1)$ — and the cosine map's clustering there is merely harmless. Two numerical guards accompany the window: q_T nodes with $(C+Q)/M \le 1$ are skipped (roundoff can push a node marginally outside the support), and inside the loop $S_2 = (Q-D)(Q+D) \le 0$ nodes are dropped for the same reason.

2.1.2 The transverse-momentum band

By (7), the window is nonempty exactly when

$$m_{aT}\, m_{bT} \;\le\; C + q_T p_T. \tag{10}$$

Both sides of (10) are strictly positive ($C = m_a E_b > 0$ and $m_{aT} m_{bT} > 0$), so squaring preserves the inequality in both directions and is an equivalence. Squaring and expanding with $m_{aT}^2 = m_a^2 + q_T^2$, $m_{bT}^2 = m_b^2 + p_T^2$:

$$\begin{aligned} \big(m_a^2 + q_T^2\big)\big(m_b^2 + p_T^2\big) \;&\le\; C^2 + 2C p_T\, q_T + q_T^2 p_T^2, \\ m_a^2 m_b^2 + m_a^2 p_T^2 + m_b^2 q_T^2 + q_T^2 p_T^2 \;&\le\; C^2 + 2C p_T\, q_T + q_T^2 p_T^2. \end{aligned} \tag{11}$$

The $q_T^2 p_T^2$ terms cancel; using $m_a^2 m_b^2 + m_a^2 p_T^2 = m_a^2 m_{bT}^2$ and collecting powers of q_T,

$$m_b^2\, q_T^2 - 2C p_T\, q_T + m_a^2 m_{bT}^2 - C^2 \;\le\; 0, \tag{12}$$

a upward parabola in q_T: the inequality holds between its roots. The quarter-discriminant is

$$\begin{aligned} \tfrac{1}{4}\Delta &= (C p_T)^2 - m_b^2\big(m_a^2 m_{bT}^2 - C^2\big) = C^2\big(p_T^2 + m_b^2\big) - m_a^2 m_b^2\, m_{bT}^2 \\ &= m_{bT}^2\big(C^2 - m_a^2 m_b^2\big) = m_{bT}^2\, m_a^2\big(E_b^2 - m_b^2\big) = m_{bT}^2\, m_a^2\, p_b^2 \;\ge\; 0, \end{aligned} \tag{13}$$

where the second line used $p_T^2 + m_b^2 = m_{bT}^2$, then $C = m_a E_b$, then $p_b^2 = E_b^2 - m_b^2$. The discriminant is a perfect square — the roots are always real, as they must be (the band exists for every open decay). The quadratic formula then gives

$$q_T^{\pm} = \frac{C p_T \pm m_a p_b\, m_{bT}}{m_b^2} = \frac{m_a\big(E_b\, p_T \pm p_b\, m_{bT}\big)}{m_b^2}, \qquad q_T \in \big[\max(0, q_T^-),\, q_T^+\big]. \tag{14}$$

The sign of the lower root follows from $\big(E_b p_T\big)^2 - \big(p_b m_{bT}\big)^2 = E_b^2 p_T^2 - p_b^2\big(m_b^2 + p_T^2\big) = p_T^2\big(E_b^2 - p_b^2\big) - p_b^2 m_b^2 = m_b^2\big(p_T^2 - p_b^2\big)$, so $q_T^- > 0 \iff p_T > p_b$. The lower root is positive iff $p_T > p_b$: a daughter harder than the rest-frame momentum can only come from a moving parent, while for $p_T < p_b$ even a parent at rest contributes and the band starts at $q_T = 0$.

The band width, $q_T^+ - q_T^- = 2 m_a p_b m_{bT}/m_b^2$, carries the m_b^{-2} of the roots: light daughters have enormous bands (for $\rho \to \pi\pi$ at $p_T = 0.6\,\mathrm{GeV}$ the band is $[0.5,\, 18]\,\mathrm{GeV}$), which is why the feed-down matrices of Sec. 5 are effectively dense. In practice the reach in q_T is limited not by the band but by the source: for a thermal-like parent spectrum $\sim e^{-(m_{aT} - m_a)/T_{\mathrm{eff}}}$ the integrand is negligible beyond

$$q_T^{\max} \;=\; \sqrt{\big(m_a + T_{\mathrm{eff}} \ln(1/\varepsilon)\big)^2 - m_a^2}, \tag{15}$$

(ε the tolerated relative tail, e.g. 10^{-8}), which also sets the required extent of any fixed parent grid. Note that T_{eff} is the slope of the *actual* source — radial flow blueshifts it well above the freeze-out temperature.

2.1.3 The logarithmic singularity at the internal band point

Inside the band the window changes topology where η_{lo} reaches zero, i.e. where $C - Q - M$ changes sign (case boundary of (8)). Solving $m_{aT} m_{bT} = C - q_T p_T$ by the same squaring as in

Sec. 2.1.2 — legitimate on the branch $C - q_T p_T \geq 0$, which holds there since $m_{aT} m_{bT} > 0$ — flips only the sign of the linear term in (12), so the roots are $-q_T^{\mp}$, and the positive one is

$$q_T^* = \frac{m_a\big(p_b\, m_{bT} - E_b\, p_T\big)}{m_b^2} = -\, q_T^-, \tag{16}$$

which lies inside the band $(0, q_T^+)$ precisely when $q_T^- < 0$, i.e. $p_T < p_b$.

For the local behavior define $g \equiv Q + M - C$, so $g < 0$ is the two-interval regime ($\eta_{\rm lo} > 0$) and $g > 0$ the merged one, with $g(q_T^*) = 0$; g is strictly increasing in q_T ($\mathrm{d}g/\mathrm{d}q_T = p_T + q_T\, m_{bT}/m_{aT} > 0$), so near the transition $g \simeq g'(q_T^*)\,(q_T - q_T^*)$. Expand the integrand factors around $\Delta\eta = 0$ using $\cosh\Delta\eta = 1 + \frac{1}{2}\Delta\eta^2 + \mathcal{O}(\Delta\eta^4)$:

$$Q + D = Q + M\cosh\Delta\eta - C \;=\; g + \tfrac{1}{2} M \Delta\eta^2 + \mathcal{O}(\Delta\eta^4), \qquad Q - D \;=\; 2Q - g - \tfrac{1}{2} M\Delta\eta^2 \;\simeq\; 2Q, \tag{17}$$

the second factor staying regular (it vanishes at the *other* edge $\eta_{\rm hi}$, far from $\Delta\eta = 0$). Hence, up to a cutoff Λ separating the local region from the rest of the window,

$$\int \frac{\mathrm{d}\Delta\eta}{\sqrt{(Q-D)(Q+D)}} \;\simeq\; \frac{1}{\sqrt{2Q}} \int \frac{\mathrm{d}\Delta\eta}{\sqrt{g + \frac{1}{2} M \Delta\eta^2}}, \tag{18}$$

and the local integral has elementary antiderivatives on both sides of the transition:

$$g > 0: \quad \int_0^{\Lambda} \frac{\mathrm{d}\Delta\eta}{\sqrt{g + \frac{1}{2} M\Delta\eta^2}} = \sqrt{\tfrac{2}{M}}\, \operatorname{arcsinh}\Big(\Lambda\sqrt{\tfrac{M}{2g}}\Big) = \sqrt{\tfrac{2}{M}}\Big[\ln\big(2\Lambda\sqrt{\tfrac{M}{2}}\big) - \tfrac{1}{2}\ln g\Big] + \mathcal{O}(g), \tag{19}$$

$$g < 0: \quad \int_{\eta_{\rm lo}}^{\Lambda} \frac{\mathrm{d}\Delta\eta}{\sqrt{\frac{1}{2} M\big(\Delta\eta^2 - \eta_{\rm lo}^2\big)}} = \sqrt{\tfrac{2}{M}}\, \operatorname{arccosh}\Big(\frac{\Lambda}{\eta_{\rm lo}}\Big) = \sqrt{\tfrac{2}{M}}\Big[\ln\big(2\Lambda\sqrt{\tfrac{M}{2}}\big) - \tfrac{1}{2}\ln|g|\Big] + \mathcal{O}(g), \tag{20}$$

where the second line used $g + \frac{1}{2} M\Delta\eta^2 = \frac{1}{2} M(\Delta\eta^2 - \eta_{\rm lo}^2)$ with $\eta_{\rm lo} = \sqrt{-2g/M}$ and $\operatorname{arccosh} x = \ln(2x) + \mathcal{O}(x^{-2})$. Both sides carry the *same* coefficient $-\frac{1}{2}\sqrt{2/M}\,\ln|g|$: the inner integral has a symmetric, integrable logarithmic singularity

$$I(q_T) \;=\; -\frac{1}{\sqrt{2QM}}\, \ln\big|q_T - q_T^*\big| \;+\; (\text{regular}), \tag{21}$$

after inserting the linearization of g. A Gauss rule straddling a logarithm converges only algebraically, so the q_T interval is split at q_T^* whenever $q_T^- < 0 < q_T^* < q_T^+$; the residual $\theta\ln\theta$ endpoint behavior after the cosine map is discussed in Sec. 2.2.1.

2.2 Numerical treatment: the cosine-mapped Gauss rule

Every non-smooth feature located above is removed by a single substitution: the $1/\sqrt{\ }$ window edges and the $\sqrt{\ }$ zeros of the μ average both cancel against a cosine map's Jacobian, leaving the lone q_T^* logarithm, which is isolated by splitting the interval there.

2.2.1 The cosine substitution and the tensor-product rule

Every one-dimensional interval $[x_{\rm lo}, x_{\rm hi}]$, of midpoint $m = \frac{1}{2}(x_{\rm lo} + x_{\rm hi})$ and half-width $h = \frac{1}{2}(x_{\rm hi} - x_{\rm lo})$, is traversed via

$$x(\theta) = m - h\cos\theta, \qquad \theta \in (0,\pi), \qquad \mathrm{d}x = h\,\sin\theta\,\mathrm{d}\theta, \tag{22}$$

which maps $\theta = 0 \mapsto x_{\rm lo}$ and $\theta = \pi \mapsto x_{\rm hi}$ monotonically. The endpoint expansion is exact in half-angle form:

$$x - x_{\rm lo} = h\,(1-\cos\theta) = 2h\sin^2\tfrac{\theta}{2}, \qquad x_{\rm hi} - x = h\,(1+\cos\theta) = 2h\cos^2\tfrac{\theta}{2}. \tag{23}$$

Consider an integrand with the inverse-square-root endpoint behavior established in Sec. 2.1.1, $f(x) = g(x)/\sqrt{(x-x_{\rm lo})(x_{\rm hi}-x)}$ with g smooth. Using (23) and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$,

$$f\big(x(\theta)\big)\,\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{g\big(x(\theta)\big)\;h\cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sqrt{2h\sin^2\frac{\theta}{2}\;\cdot\;2h\cos^2\frac{\theta}{2}}} = g\big(x(\theta)\big)\,: \tag{24}$$

the singular factors cancel *identically*, not just to leading order — the θ-integrand is as smooth as g, and Gauss–Legendre in θ converges spectrally. (If only one endpoint is singular the cancellation happens there and the other endpoint merely gains harmless node clustering; endpoint zeros $\propto \sqrt{x - x_{\rm lo}}$, as in the μ integral, become smooth $\propto \sin^2\frac{\theta}{2}$ by the same algebra.)

The θ integral itself is discretized by an n-point Gauss–Legendre rule with nodes $x_i \in (-1,1)$ and weights w_i, mapped affinely to $(0,\pi)$: $\theta_i = \frac{\pi}{2}(x_i+1)$, $\int_0^\pi G\,\mathrm{d}\theta \approx \frac{\pi}{2}\sum_i w_i G(\theta_i)$. Combining with (22),

$$\int_{x_{\rm lo}}^{x_{\rm hi}} f(x)\,\mathrm{d}x \;\approx\; h\sum_{i=1}^{n} v_i\, f\big(m - h\,c_i\big), \qquad c_i = \cos\theta_i, \quad v_i = \tfrac{\pi}{2}\,w_i\sin\theta_i, \tag{25}$$

so the map costs no trigonometry at evaluation time: the fused pairs (c_i, v_i) are computed once per order and shared (memoized) across all decay objects. Applying (25) in both directions gives the tensor-product kernel sum

$$K(E_b) \;\approx\; \sum_{\rm seg}\; h_q \sum_{i=1}^{n_q} v_i^q\; q_T^{(i)}\; h_\eta\Big(q_T^{(i)}\Big) \sum_{j=1}^{n_\theta} \frac{v_j^\theta}{\sqrt{Q^2 - D_{ij}^2}}\,\big[\text{bracket}\big]_{ij}, \tag{26}$$

where the inner $1/\sqrt{\ }$ is evaluated in the factored form $(Q-D)(Q+D)$ and the singular endpoint behavior is tamed by (24) through the mapped $\Delta\eta$ nodes. The only residual non-smoothness is the logarithm of Sec. 2.1.3 at the segment endpoint q_T^*: with $x - q_T^* \propto \theta^2$ from (23), the integrand behaves as $\sin\theta\cdot\ln\theta^2 \sim 2\,\theta\ln\theta$ near $\theta = 0$ — continuous but not analytic — for which the Gauss error decreases algebraically, $\sim n_q^{-4}$ (measured: the error at low p_T drops $\approx 16\times$ per doubling of n_q).

2.2.2 Three-body: the effective-mass average

For $a \to b + c + d$ the kernel is averaged over the effective mass $\mu \in [m_c + m_d,\, m_a - m_b]$ with weight $p_c^{\tilde{c}}(\mu)$ and normalization $\mathcal{N} = \int \mathrm{d}\mu\, p_b^a(\mu)\, p_c^{\tilde{c}}(\mu)$ [Eqs. (45)–(48)]. The integrand vanishes as a square root at both endpoints: $p_c^{\tilde{c}} \sim \sqrt{\mu - (m_c + m_d)}$ at threshold, and $K \propto$ band width $\propto p_b^a(\mu) \sim \sqrt{m_a - m_b - \mu}$ at the upper limit. The same substitution (22) turns $\sqrt{\ }$-type zeros into smooth θ^2 behavior, and the μ integral is done with an n_μ-point cosine-mapped Gauss rule; the nodes μ_i, energies $E_b(\mu_i)$ and normalized weights are precomputed at construction.

2.2.3 Measured convergence and default orders

For a Boltzmann source ($T = 0.15\,\mathrm{GeV}$) relative errors against high-order references behave as follows (worst case is low p_T, where the q_T^* logarithm is inside the band):

orders		error at $p_T = 0.2$	error at $p_T = 0.8$
$\rho \to \pi\pi$:	$(n_q, n_\theta) = (128, 32)$	$7 \cdot 10^{-8}$	$5 \cdot 10^{-12}$
	$(64, 16)$	$2 \cdot 10^{-6}$	$5 \cdot 10^{-12}$
	$(32, 8)$	$3 \cdot 10^{-6}$	$2 \cdot 10^{-8}$
$\omega \to 3\pi$:	$(n_\mu, n_q, n_\theta) = (32, 128, 32)$	$2 \cdot 10^{-8}$	$3 \cdot 10^{-13}$
	$(16, 64, 16)$	$5 \cdot 10^{-7}$	$3 \cdot 10^{-10}$
	$(8, 32, 8)$	$1 \cdot 10^{-4}$	$5 \cdot 10^{-5}$

The $\Delta\eta$ direction is spectrally convergent and over-resolved already at $n_\theta = 16$; n_q is limited by the logarithm ($\sim n_q^{-4}$, only relevant for $p_T < p_b$); $n_\mu \geq 16$ is safe while $n_\mu = 4$ fails at the percent level. The exported functor constructors default to $(n_q, n_\theta, n_\mu) = (128, 32, 32)$; these settings are conservative, and cost scales linearly in each order. The production `CascadeTable` instead defaults to $(n_q, n_\theta, n_\mu, n_\sigma, n_{m_a}) = (32, 8, 8, 8, 8)$, trading some build accuracy for speed on its sampled grid. The continuous-functor regression tests give number conservation, $N_b = B\,N_a$, at the 10^{-4} level or better, but they use the higher exported-functor defaults; they do not by themselves certify the production table orders.

3 Multi-body decays: the effective-mass average

The three-body μ average of Sec. 2 is the first instance of a general pattern: any number of unobserved daughters can be clustered into a single effective particle of mass μ, and the same two-body kernel is averaged over that cluster's mass. This section makes the pattern explicit — its *analytic* form is a phase-space recursion, its *numerical* cost is one extra mass quadrature per additional daughter — and applies it to four-body decays. (Finite widths, Sec. 4, are the very same construction with a spectral function in place of the phase-space weight.) The organizing question throughout is whether the extra integration multiplies the number of *expensive* kernel evaluations or only enriches the *cheap* weight — and here, as we will see, it is only the weight.

3.1 Recursive effective-mass structure

Return to the three-body average and read it structurally. For $a \xrightarrow{\sim} b + (cd)$ the observed daughter b recoils against the pair treated as a single effective particle $\widetilde{cd}$ of mass μ; the *angular* distribution of b — hence the entire $(q_T, \Delta\eta)$ kernel — depends on the pair only through μ, via $E_b(\mu) = (m_a^2 + m_b^2 - \mu^2)/2m_a$. The internal breakup of the pair enters solely through *how much probability sits at each* μ, i.e. the invariant-mass density of the pair,

$$\rho_2(\mu; m_c, m_d) \;\propto\; q(\mu; m_c, m_d), \qquad q(M; m_1, m_2) \equiv \frac{\lambda^{1/2}(M^2, m_1^2, m_2^2)}{2M}, \tag{27}$$

which is precisely the weight $p_c^{\tilde{c}}(\mu)$ of Eq. (47). Writing the map this way,

$$\mathcal{F}_b = \frac{B\,m_a}{4\pi\,\mathcal{N}} \int_{\mu_-}^{\mu_+} \mathrm{d}\mu\; \underbrace{p_b^a(\mu)\,\rho_2(\mu)}_{\text{weight}}\; K\big(E_b(\mu)\big), \qquad \mathcal{N} = \int \mathrm{d}\mu\, p_b^a(\mu)\,\rho_2(\mu), \tag{28}$$

makes the generalization immediate. For an n-body decay $a \to b + (c_1 \cdots c_{n-1})$, cluster the $n-1$ unobserved particles into $\widetilde{c}$ of mass $\mu \in [\sum_i m_{c_i},\, m_a - m_b]$; the kernel is unchanged, $K(E_b(\mu))$, and only the weight generalizes to the invariant-mass density $\rho_{n-1}(\mu)$ of the cluster.

Where the density comes from. The density and its recursion are the standard factorization of n-body Lorentz-invariant phase space. Peeling the observed particle b against the cluster of invariant mass μ,

$$\mathrm{d}\Phi_n\big(m_a^2;\, p_b, \{p_{c_i}\}\big) = \frac{\mathrm{d}\mu^2}{2\pi}\, \mathrm{d}\Phi_2\big(m_a^2;\, m_b^2, \mu^2\big)\, \mathrm{d}\Phi_{n-1}\big(\mu^2;\, \{p_{c_i}\}\big), \tag{29}$$

and integrating the two-body factor over the isotropic emission angle, $\int \mathrm{d}\Phi_2(m_a^2; m_b^2, \mu^2) \propto q(m_a; m_b, \mu) = p_b^a(\mu)$, identifies the μ-measure as $p_b^a(\mu)\,\mathrm{d}\mu$ times the cluster's own phase-space volume. Defining that volume as the density, $\rho_{n-1}(\mu)\,\mathrm{d}\mu \equiv \int \mathrm{d}\Phi_{n-1}(\mu^2; \{p_{c_i}\})$ at fixed cluster mass, and applying (29) once more to $\mathrm{d}\Phi_{n-1}$ — peeling c_1 against the sub-cluster of mass σ — gives the Byckling–Kajantie recursion, one particle at a time,

$$\rho_k\big(\mu; m_{c_1}, \ldots, m_{c_k}\big) = \int_{\sigma_-}^{\sigma_+} \mathrm{d}\sigma\; q(\mu; m_{c_1}, \sigma)\; \rho_{k-1}\big(\sigma; m_{c_2}, \ldots, m_{c_k}\big), \qquad \rho_1(\sigma; m) = \delta(\sigma - m), \tag{30}$$

with $\sigma \in [\sum_{i\geq 2} m_{c_i},\, \mu - m_{c_1}]$. The overall constants (2π from (29), the 4π from each angular integration) are μ-independent and cancel in the normalized ratio $\mathcal{F}_b$, so they are dropped. The recursion terminates at $\rho_1 = \delta$ (a single particle has definite mass); one step gives $\rho_2 = q$ (the three-body weight of Eq. 47), and for four-body

$$\rho_3(\mu; m_c, m_d, m_e) = \int_{m_d+m_e}^{\mu - m_c} \mathrm{d}\sigma\; q(\mu; m_c, \sigma)\; q(\sigma; m_d, m_e). \tag{31}$$

The four-body map explicitly. Assembling the pieces, the observed-b spectrum of $a \to b\,c\,d\,e$ is the two-dimensional mass average

$$\begin{aligned} \mathcal{F}_b &= \frac{B\, m_a}{4\pi\, \mathcal{N}} \int_{m_c+m_d+m_e}^{m_a - m_b} \mathrm{d}\mu\; p_b^a(\mu) \\ &\quad \times \underbrace{\left[\int_{m_d+m_e}^{\mu - m_c} \mathrm{d}\sigma\; q(\mu; m_c, \sigma)\, q(\sigma; m_d, m_e)\right]}_{\rho_3(\mu)} K\big(E_b(\mu)\big), \\ \mathcal{N} &= \int \mathrm{d}\mu\, p_b^a(\mu) \rho_3(\mu). \end{aligned} \tag{32}$$

The bracket is the Lorentz-invariant phase-space volume of the three-particle cluster $c\,d\,e$ at total mass μ — geometrically, the area of its Dalitz plot. The inner σ integral runs over the (de) sub-mass; by the symmetry of phase space any pairing gives the same ρ_3, so the choice of which particle to peel first is free. Both mass integrals carry the $\sqrt{\ }$-type endpoint zeros of the Källén function (each q vanishes at its threshold), regularized by the cosine map of Sec. 2.2.1.

Cost. The crucial point is that $\rho_{n-1}(\mu)$ is *independent of the kernel*: it is a static, $(n-2)$-fold nested phase-space integral in the mass variables alone, computed once on the n_μ nodes

at construction. The number of expensive kernel evaluations therefore stays n_μ — the *same* as three-body — for any number of unobserved daughters; only the weight becomes a nested (but kernel-free, hence cheap) integral. A four-body decay matrix costs essentially what a three-body one costs. The inner mass integrals inherit the same $\sqrt{\ }$ endpoint zeros as the μ integral (the Källén function vanishes at each threshold and pseudo-threshold), and are handled by the same cosine-mapped Gauss rule of Sec. 2.2.1. For general n the weight is an $(n-2)$-fold nested mass integral; with n_σ points per level it costs $n_\mu\, n_\sigma^{n-2}$ evaluations of the elementary function q — arithmetic only, no kernel — against the fixed n_μ kernel evaluations. So the build cost is dominated by the kernel until n is large, and a four- or five-body matrix is, in practice, a three-body matrix with a slightly richer weight.

On observing a cluster particle. In the cascade every daughter is observed in turn (Sec. 5). When the observed particle is one of the "cluster" members, one simply relabels: the observed particle plays the role of b and the remaining $n-1$ form the cluster. The construction above is symmetric under that relabelling, so nothing new is required — the observed slot is chosen, the rest are clustered, and ρ_{n-1} is built from their masses.

4 Finite widths

So far every mass has been sharp. A resonance of width Γ_i is instead spread over a spectral function — a relativistic Breit–Wigner,

$$\varrho_i(m) \;\propto\; \frac{m^2\,\Gamma_i(m)}{\big(m^2-m_i^2\big)^2+m_i^2\,\Gamma_i(m)^2}, \qquad \int \varrho_i(m)\,\mathrm{d}m = 1, \tag{33}$$

whose width is itself mass dependent when the resonance decays to two particles in relative angular momentum ℓ,

$$\Gamma_i(m) = \Gamma_i\,\frac{m_i}{m}\left(\frac{q(m;m_1,m_2)}{q(m_i;m_1,m_2)}\right)^{2\ell+1}\frac{B_\ell\big(q(m)\big)}{B_\ell\big(q(m_i)\big)} \tag{34}$$

(B_ℓ the Blatt–Weisskopf barrier factor). It vanishes at threshold $m = m_1+m_2$, so ϱ_i carries no weight below threshold, and $\varrho_i \to \delta(m-m_i)$ as $\Gamma_i \to 0$.

Equations above describe the general mass-dependent-width construction. Both specializations are implemented, as subtypes of `SpectralShape`, and selected by the `spectral` table option (`CascadeTable`) or the `shape` keyword (`ThermalModel`). Each returns an *unnormalized* shape on a caller-specified window; `BroadParent` normalizes when forming the parent-mass average.

`MassDependentBW` is the default. It implements (34) at $\ell = 0$ and *without* the Blatt–Weisskopf factor, with the symmetric split $q(m) = \sqrt{m^2 - m_{\mathrm{th}}^2}/2$, where m_{th} is the summed daughter mass of the resonance's *lightest* channel — so the shape is a property of the species, while each individual channel's closure remains handled by dropping parent-mass nodes below its own threshold. The split is exact for $\sigma \to \pi\pi$ and approximates the q-ratio for asymmetric pairs. Since $\Gamma(m)$ vanishes at m_{th}, the shape carries no weight where the state cannot decay; for the σ that removes the 2.56% of constant-width weight below $2m_\pi$. A resonance whose pole lies at or below its own threshold has no $q(m_i)$ and falls back to the constant shape.

`BreitWigner` is the constant-width specialization $\Gamma_i(m) = \Gamma_i$, retained both as that fallback and as the `:constant` option. Barrier factors and per-channel ℓ remain unimplemented.

One principle, two costs. Whether a given width is cheap or expensive to include is decided by *how that mass enters the decay map.* A mass that enters *only through the effective cluster mass* μ — any of the unobserved daughters — reshapes the precomputed weight $\rho_{n-1}(\mu)$ and nothing else, so it costs nothing extra. A mass that enters the *kinematics of the observed leg* — the parent (through $C = m_a E_b$ and the q_T band) or the observed daughter (through E_b, p_b) — changes the kernel itself, and must be handled by an *outer average*: one kernel build per mass node. The three places a width can sit map onto exactly this split:

- **Unobserved / cluster masses — cheap.** A width on any c_i just averages the sharp momentum inside the phase-space recursion (30), $q(\mu; m_{c_1}, \sigma) \to \int \mathrm{d}m_{c_1}\, \varrho(m_{c_1})\, q(\mu; m_{c_1}, \sigma)$. It modifies $\rho_{n-1}(\mu)$ and touches no kernel.
- **Observed daughter b — expensive, but usually moot.** $K \to \int \mathrm{d}m_b\, \varrho_b(m_b)\, K$. The measured π, K, p, Λ are stable or narrow, so this collapses to the sharp δ; it only matters for a broad observed resonance mid-cascade.
- **Parent a — expensive, and dominant.** $\int \mathrm{d}m_a\, \varrho_a(m_a)\, (\cdots)$. The broadest states $(\sigma, \rho, \Delta, \dots)$ are parents, so this is where finite width matters most.

Which case this cascade needs. The split above is general, but the cascade of Sec. 6 decays every resonance in *its own* feeder-before-fed step (mass-descending for an ordinary pole-open graph): $a \to bR$ and $R \to cc'$ are two separate two-body decays, never collapsed into one three-body $a \to bcc'$. A broad R therefore enters *as a parent* — when R itself decays — not as a cluster mass. The cluster case is the *collapsed* picture, which this cascade does not use; and the genuine multi-body channels it does treat by the effective-mass integral ($\omega \to 3\pi$, $\eta \to 3\pi$) have *stable* daughters, carrying no cluster width either. So for this cascade the parent average — the *expensive* case — is the main use of the per-species widths in the decay list. The cheap case is kept for completeness (a collapsed treatment, or a genuinely broad unobserved daughter), but it is the exception here. The next two subsections treat each in turn, cheap first.

4.1 The cheap case: width reweights the effective-mass integral

Case 1 is not a new mechanism — it *is* the effective-mass integral of Sec. 3.1. When the unobserved pair $c\,c'$ forms a resonance R, its density $\rho_2(\mu) = q(\mu; m_c, m_{c'})$ picks up the R propagator and *becomes* a Breit–Wigner peak in μ. The two statements

$$\text{“}a \to b\,R\text{, then } R \to c\,c' \text{ with } R \text{ broad”} \quad \Longleftrightarrow \quad \text{“}a \to b\,c\,c'\text{, pair weighted by } \varrho_R(\mu)\text{”}$$

are literally the same integral (28). A narrow R collapses $\varrho_R \to \delta(\mu - m_R)$ and factorizes it back into the sharp two-body steps $a \to bR$ then $R \to cc'$; a broad R smears the two together. This is the elegant limit — the width lives entirely in a weight the effective-mass integral already computes, a change to the density rather than the kernel. But it is exactly the *collapsed* step the sequential cascade avoids: there the same broad R reappears as a parent in its own decay (next subsection), which is where its width acts. So this route, though cheap, is the exception in this cascade, not the rule.

4.2 The main case: parent width as a matrix average

For the parent (or observed daughter) the mass reshapes the kinematics, so there is no weight to hide it in. But the matrix form (Sec. 5) keeps it clean: the sharp-mass map is linear in

the source and m_a enters only kinematically, so the width average is a weighted sum of sharp matrices,

$$W_n^{\text{eff}} = \int \mathrm{d}m_a\, \varrho_a(m_a)\, W_n(m_a) \;\approx\; \sum_i w_i\, W_n\big(m_a^{(i)}\big), \tag{35}$$

built at n_{m_a} mass nodes across the support of ϱ_a. The result is a matrix of the *same shape* — cascade, apply and persistence are untouched, and the width is absorbed once, at build time. (Interpolating W_n in m_a instead of rebuilding fails: the q_T band $[q_T^-, q_T^+]$ of (14) *moves* with m_a, so the matrices have shifting support and do not interpolate.) Two consequences follow.

A width opens closed channels. At the sharp pole mass a channel with $\sum_{\text{daughters}} m > m_a$ is forbidden, and the cascade skips it. But ϱ_a has support above the pole: on the tail $m_a' > \sum m$ the daughters fit, and the channel contributes. Nothing special is required — the same mass average (35) does it. Numerically, the normalization spans the full spectral window, while kernel nodes are placed only in its open part; the closed region therefore contributes spectral normalization but requires no kernel evaluations:

$$W_n^{\text{eff}}\big|_{\text{closed channel}} = \int_{\sum m}^{m_{\max}} \mathrm{d}m_a'\; \varrho_a(m_a')\, W_n(m_a'). \tag{36}$$

A channel that is exactly zero in the sharp limit thus acquires a small weight from the parent's high-mass tail — this is the physical meaning of the "closed-at-nominal-mass" count reported by list validation and channel enrollment.

Production and decay use the same ϱ_a. In a thermal source the parent is produced already dressed with its spectral function, so $\mathcal{F}_a$ carries ϱ_a; the decay then needs the parent's mass *conditional* on its momentum. The standard approximation uses the same ϱ_a in both places — exact as $\Gamma_a \to 0$, and controlled otherwise. The factorized form here ($\mathcal{F}_a$ times the ϱ_a-averaged decay) is the usual on-shell-spectral compromise, good while Γ_a/m_a is not large.

Status. Four-body is implemented (the 532 cluster keys of Sec. 3.1), and so now is the *parent width* — the main case for this cascade. Each species with $\Gamma > 1\,\text{MeV}$ is averaged over its (constant-width, truncated) Breit–Wigner by (35): the decay matrix at n_{m_a} parent-mass nodes, weighted by ϱ_a and summed, giving a same-shape matrix that the cascade uses unchanged. The mass integral uses the tan-substitution $s = m_R^2 + m_R\Gamma \tan\psi$, with node mass $m_a = \sqrt{s}$, which flattens the Lorentzian so plain Gauss–Legendre in ψ resolves the peak for any Γ (a cosine map would starve it). When threshold cuts the window, a cosine map is used on the open interval to absorb the integrable $1/\sqrt{m_a - \sum m}$ endpoint behavior. The full window still sets the normalization, so a channel closed at the pole keeps its open-tail fraction — channel reopening, for free. Validated by number conservation, the single-node $\to$ sharp and narrow-Γ $\to$sharp limits, spectral convergence in n_{m_a}, and reopening. In the full-list benchmark that motivated this implementation, roughly 3,200 of about 18,000 keys were broad, shifting the feed-down at the few-percent level. The averaged matrix has the *same per-key shape*, so width does not multiply its stored size by n_{m_a}; it can nevertheless create additional unique keys because Γ is part of (54). Width mainly multiplies build work by n_{m_a} for each broad key, so persistence (Sec. 6) is essential. The cheap cluster reweight (for a genuine broad unobserved daughter) and observed-daughter width (rare) remain future work.

One kernel average. The effective-mass average of Sec. 3 and the parent-width average above are the *same* object: a coef-weighted sum of the two-body kernel over mass nodes,

$$\mathcal{F}_b = \sum_i c_i \, K\big(\mathcal{F}_a;\; m_a^{(i)},\, m_b,\, E_b^{(i)}\big), \tag{37}$$

differing only in *which* mass is swept and how c_i is built. Two-body is a single node. The effective-mass average sweeps the *cluster* mass μ at fixed m_a, with $E_b^{(i)} = E_b(\mu_i)$ and $c_i \propto \rho_{n-1}(\mu_i)$. The parent-width average sweeps the *parent* mass m_a, with $c_i \propto \varrho_a(m_a^{(i)})$ — and a broad multi-body parent is simply the two nested: its parent-mass nodes × cluster nodes flatten into one list. All three are therefore the *same* node sum, and the matrix assembly (Sec. 5) — which reads only the per-node $(m_a^{(i)}, E_b^{(i)}, c_i)$ — consumes them identically, distinguishing nothing.

4.3 What the width shape controls, and what it does not

Two properties of ϱ_i are frequently conflated: its behavior at threshold, fixed by the mass dependence of $\Gamma_i(m)$, and the weight in its high-mass tail, fixed by the truncation window. They are independent, and only the first is repaired by (34).

For a two-body S-wave decay ($\ell = 0$, $B_0 \equiv 1$) (34) reduces to

$$\Gamma_i(m) = \Gamma_i \, \frac{m_i}{m} \, \frac{q(m)}{q(m_i)}, \qquad q(m) = \tfrac{1}{2}\sqrt{m^2 - 4m_\pi^2} \;\text{ for } i \to \pi\pi, \tag{38}$$

which vanishes at $m = 2m_\pi$ as promised. Its large-m behavior is the part that surprises: $q(m) \to m/2$ cancels the m_i/m prefactor, so

$$\Gamma_i(m) \;\xrightarrow[m\to\infty]{}\; \Gamma_i \, \frac{m_i}{2\,q(m_i)} \;\equiv\; \Gamma_i^\infty, \tag{39}$$

a *constant*, and for a broad low-mass state Γ_i^∞ exceeds Γ_i. Since $\varrho_i \sim \Gamma_i(m)/m^2$ at large m either way, the S-wave mass-dependent width leaves the tail exponent unchanged and mildly *increases* its coefficient. Only for $\ell \geq 1$, where $q^{2\ell+1}$ outruns the propagator, does (34) suppress the tail.

Take the σ as the extreme case: $m_i = 0.475$, $\Gamma_i = 0.550$, $q(m_i) = 0.192$, hence $\Gamma_i^\infty = 0.680 > \Gamma_i$. With the window $[\, m_i - k\Gamma_i,\; m_i + k\Gamma_i \,]$ at $k = 4$ the weight distributes as

weight in	constant Γ_i	$\Gamma_i(m)$ of (38)
below $2m_\pi$	2.56%	0.00%
0.279–1.0 GeV	69.85%	68.53%
1.0–2.0 GeV	22.73%	25.78%
2.0–2.675 GeV	4.86%	5.69%
peak of ϱ_i	0.588 GeV	0.592 GeV

The mass dependence removes the sub-threshold weight exactly and moves the peak by 4 MeV; the tail grows slightly. This matters because of the normalization convention of Sec. 4: the *full* window normalizes ϱ_i while only its open part carries decay nodes. That is the correct choice for channel reopening — a channel closed at the pole keeps its open-tail fraction — but with a constant width it also means the 2.56% lying below $2m_\pi$, where no decay exists, is normalized in and emits nothing. A state parametrized this way yields slightly fewer than two pions per decay.

The tail, by contrast, is governed entirely by k:

weight above 1 GeV	$k=1$	$k=2$	$k=4$	$k=8$
constant Γ_i	1.7%	19.2%	27.6%	31.8%
$\Gamma_i(m)$	2.0%	22.1%	31.5%	36.1%

A factor of sixteen in the hard-daughter source, set by a truncation constant rather than by (m_i, Γ_i). For narrow states this is irrelevant — the window closes long before anything interesting — but for $\Gamma_i \sim m_i$ it is the dominant modelling choice, and it should be recognized as one. Where a broad state's tail overlaps other entries of the decay list that carry the same quantum numbers, the tail also double counts them.

The practical consequence is a division of labour: use (34) to fix threshold behavior, and a support cap or a form factor — not the mass dependence — to control the tail.

4.4 A spectral-function interface

Nothing downstream of the parent-mass average inspects the shape of ϱ_i. `BroadParent` consumes only the triple

$$\big(\{m_a^{(i)}\},\ \{w_i\},\ \mathcal{N}\big)$$

— nodes, weights, and the full-support normalization — and (37) consumes only $(m_a^{(i)}, E_b^{(i)}, c_i)$. A spectral function is therefore fixed by a two-method contract: evaluation $\varrho(m)$ on its support, and a quadrature rule returning that triple for a given threshold. Any shape satisfying it — constant-width Breit–Wigner, (34), a form-factor-damped variant, or a tabulated $\mathrm{d}\delta_I^\ell/\mathrm{d}m$ from measured phase shifts — drops in without touching the kernel, the matrix assembly, or the cascade.

The quadrature is the only method with any subtlety. The tan-substitution $s = m_i^2 + m_i\Gamma_i \tan\psi$ of Sec. 4 flattens the Lorentzian *exactly* for a constant width, and is merely a good change of variables for a mass-dependent one: the map is built from the pole parameters while the integrand carries the true ϱ_i, so nodes cluster where the weight is, and correctness rests on the quadrature rather than on the map being exact. For the σ the peak moves by 4 MeV between the two shapes, so the same nodes serve both.

5 The matrix representation: one matrix per mode

Let the parent mode $\mathcal{F}_{a,n}$ be independent of η_q and sampled on a grid $q_T^{(1)}, \ldots, q_T^{(N_q)}$, and let interpolation reconstruct it anywhere via local Lagrange cardinal weights L_j of order P (stencil of P points, error $\mathcal{O}(h^P)$),

$$\mathcal{F}_{a,n}(q_T) \;\approx\; \sum_j L_j(q_T)\, \mathcal{F}_{a,n}\big(q_T^{(j)}\big), \qquad L_j\big(q_T^{(i)}\big) = \delta_{ij}. \tag{40}$$

First the mode bookkeeping. Insert the mode input $\mathcal{F}_a(q_T, \eta_q, \phi_q) = \mathcal{F}_{a,n}(q_T)\cos(n\phi_q)$ into the source bracket of (1), evaluated at $\phi_p = 0$; together with the two rapidity signs (equal contributions, since neither the kernel nor this source distinguishes $\pm\Delta\eta$) the four source evaluations are

$$\begin{aligned}\sum_{s=\pm}\sum_{r=\pm} \mathcal{F}_{a,n}(q_T)\cos\big(n(0+r\alpha)\big) &= \mathcal{F}_{a,n}(q_T)\,\big[2\cos(n\alpha) + 2\cos(-n\alpha)\big] \\ &= 4\cos(n\alpha)\,\mathcal{F}_{a,n}(q_T) = 4\,T_n(Z)\,\mathcal{F}_{a,n}(q_T),\end{aligned} \tag{41}$$

using the evenness of the cosine and $\cos(n \arccos Z) = T_n(Z)$ [Eq. (40)]. Now discretize (1) by the tensor rule (26) and insert the interpolation (40) at every q_T node:

$$\mathcal{F}_{b,n}\big(p_T^{(k)}\big) \approx \frac{B\, m_a}{4\pi p_b} \sum_{\text{seg}} h_q \sum_i v_i^q\, q_T^{(i)}\, h_\eta^{(i)} \sum_{j'} \frac{v_{j'}^\theta}{\sqrt{Q^2 - D_{ij'}^2}}\, 4T_n(Z_{ij'}) \underbrace{\sum_j L_j\big(q_T^{(i)}\big)\, \mathcal{F}_{a,n}\big(q_T^{(j)}\big)}_{\text{interpolated source}}$$

$$= \sum_{j=1}^{N_q} \underbrace{\left[\frac{B\, m_a}{4\pi p_b} \sum_{\text{seg}} h_q \sum_i v_i^q\, q_T^{(i)}\, h_\eta^{(i)} \sum_{j'} \frac{v_{j'}^\theta\, 4T_n(Z_{ij'})}{\sqrt{Q^2 - D_{ij'}^2}}\, L_j\big(q_T^{(i)}\big) \right]}_{= W_n[k,j]} \mathcal{F}_{a,n}\big(q_T^{(j)}\big), \tag{42}$$

where the second line only exchanged the finite sums — legitimate because every factor except $\mathcal{F}_{a,n}$ is independent of the source. The matrix element $W_n[k,j]$ is thus the exact quadrature of the kernel against the j-th cardinal function. Assembly therefore runs the *same* quadrature loops as a functor evaluation; wherever the source would be evaluated, the bracketed multiplier is instead scattered onto the P columns of the interpolation stencil of $q_T^{(i)}$ ("scatter assembly"). One row costs one kernel evaluation with no source calls; the whole matrix costs N_{p_T} of them. Equivalently — and this was used as a cross-check — column j of W_n equals the functor applied to the j-th cardinal basis function, by the same linearity; the scatter form merely avoids the N_q-fold redundancy of computing columns one at a time.

Two properties of the quadrature matter for the matrix:

- The Θ-support in q_T is the band $q_T^\pm = m_a\big(E_{b|c}^a\, p_T \pm p_{b|c}^a\, m_{bT}\big)/m_b^2$; each row is supported (up to the stencil halo) inside its band. For light daughters the band is wide ($\propto m_b^{-2}$), so in practice the $\rho \to \pi\pi$ and $\omega \to \pi\pi\pi$ examples are effectively dense (measured fill 65–76%) and numerically full rank. They are not, however, representative of every production key. A full census of the 18,299 production keys found 33.583% exact nonzeros. The contiguous support spans occupied 44.234% of dense storage and were 75.921% filled internally. `CascadeTable` now stores each row's contiguous span in an offset-backed `RowBandMatrix`: one start column per row, one sentinel offset vector, and one flat `Float32` value payload. At $G = 134$ the full packed payload is estimated at 1.1012 GiB, or 44.98% of the 2.4481 GiB dense baseline. On 128 representative real matrices, packed storage was 44.56% of dense and the packed `mul!` kernel was 1.70 times faster, with zero steady-state allocations and a maximum normalized difference of 1.177×10^{-15}.

- The integrand has an integrable $\ln|q_T - q_T^*|$ singularity at $q_T^* = -q_T^-$, where the $\Delta\eta$ domain splits; the interval is divided there. All $1/\sqrt{\ }$ endpoint singularities (in $\Delta\eta$, at $Z = \pm 1$) and $\sqrt{\ }$-type endpoint zeros (in μ) are absorbed by the substitution $x = \text{mid} - \text{half}\cos\theta$, whose $\sin\theta$ Jacobian regularizes them; fixed-order Gauss–Legendre then converges spectrally except for the residual $\theta \ln\theta$ behavior at q_T^* ($\sim n_q^{-4}$).

The only error introduced relative to the functor is interpolation, $\mathcal{O}(h^P)$; with a cubic stencil ($P = 4$) on $h = 0.035\,\text{GeV}$ a thermal source reproduces the functor to a few $\times 10^{-5}$, and the truncation "$\mathcal{F}_a \equiv 0$ outside the grid" requires the source to be negligible beyond the last grid point.

5.1 The Z-space form of the assembly integral

For the matrix assembly the inner rapidity integral can be freed of all transcendental functions. The observation is that an η-independent source never requires the rapidity $\Delta\eta$ itself — only $Z = D/Q$ enters, through the Chebyshev factor $T_n(Z)$.

5.1.1 Derivation

Start from the inner integral of the mode-n assembly at fixed q_T, with both rapidity signs folded in (factor 2, legitimate because the kernel is even in $\Delta\eta$ and the source does not depend on it) and both azimuthal roots summed (the $2T_n$ of Eq. 41):

$$I_n \;=\; 2\int_{\eta_{\rm lo}}^{\eta_{\rm hi}} \frac{\mathrm{d}\Delta\eta}{\sqrt{Q^2 - D^2}}\, 2\,T_n(Z), \qquad D = M\cosh\Delta\eta - C, \quad Z = \frac{D}{Q}. \tag{43}$$

Substitute $c = \cosh\Delta\eta$, $\mathrm{d}\Delta\eta = \mathrm{d}c/\sqrt{c^2-1}$, which maps the window (8) to $c \in [\,\max(1, (C-Q)/M),\, (C+Q)/M\,]$; then pass to

$$Z = \frac{Mc - C}{Q}, \qquad c(Z) = \frac{C + QZ}{M}, \qquad \mathrm{d}c = \frac{Q}{M}\,\mathrm{d}Z, \qquad \sqrt{Q^2 - D^2} = Q\sqrt{1 - Z^2}. \tag{44}$$

The window endpoints become $Z = 1$ (kinematic edge, $c = (C+Q)/M$) and $Z = Z_{\rm lo}$ with

$$Z_{\rm lo} = \max\!\left(-1,\; \frac{M - C}{Q}\right), \tag{45}$$

the value $(M-C)/Q$ corresponding to $c = 1$, i.e. $\Delta\eta = 0$. Collecting the Jacobians,

$$I_n = \frac{4}{M}\int_{Z_{\rm lo}}^{1} \frac{T_n(Z)\,\mathrm{d}Z}{\sqrt{(1 - Z^2)\,(c(Z)^2 - 1)}}. \tag{46}$$

This is an exact change of variables, not an approximation: the $\Delta\eta$ form of Sec. 2 and (46) are the same number. What has changed is the computational content of one quadrature node: $c(Z)$ is a multiply–add, $T_n(Z)$ follows from the recurrence $T_{k+1} = 2ZT_k - T_{k-1}$ [Eq. (40)], and the only remaining special function is one square root. No arccosh (window bounds), cosh (per node) or arccos (azimuthal angle) appears.

5.1.2 Quadrature: two regimes

Full window, $Z_{\rm lo} = -1$ (i.e. $C - Q \geq M$). The factor $1/\sqrt{1-Z^2}$ is exactly the Chebyshev–Gauss weight, so with the standard nodes the weight is integrated *exactly*:

$$I_n \;\approx\; \frac{4}{M}\,\frac{\pi}{n_\theta}\sum_{k=1}^{n_\theta} \frac{T_n(Z_k)}{\sqrt{c(Z_k)^2 - 1}}, \qquad Z_k = \cos\theta_k, \quad \theta_k = \frac{(2k-1)\pi}{2n_\theta}. \tag{47}$$

The nodes are precomputed once per order; note also $T_n(Z_k) = \cos(n\theta_k)$, so for fixed n even the Chebyshev factors are constants of the rule.

Partial window, $-1 < Z_{\rm lo} < 1$ (i.e. $C - Q < M$). Both endpoint factors vanish like square roots *inside* the domain: $1 - Z^2$ at $Z = 1$, and $c^2 - 1$ at $Z = Z_{\rm lo}$. The decisive point is that $c - 1 = \frac{Q}{M}(Z - Z_{\rm lo})$ is *exactly linear* in Z (immediate from $c(Z) = (C + QZ)/M$ and $C + QZ_{\rm lo} = M$), so the full radicand splits into an endpoint-singular part and a manifestly regular remainder,

$$\big(1 - Z^2\big)\big(c^2 - 1\big) = \underbrace{(1 - Z)(Z - Z_{\rm lo})}_{\text{endpoint zeros}} \underbrace{\tfrac{Q}{M}\,(1 + Z)\,(c + 1)}_{\equiv\, R(Z)^2}, \tag{48}$$

with $R(Z)^2$ *strictly positive on the closed interval* ($Q, M > 0$; $1 + Z > 0$ because $Z \geq Z_{\rm lo} > -1$; $c + 1 \geq 2$). The integrand is therefore exactly the form of Sec. 2.2.1, $f(Z) = g(Z)/\sqrt{(Z - Z_{\rm lo})(1 - Z)}$ with the *smooth* numerator

$$g(Z) = \frac{T_n(Z)}{R(Z)} = \frac{T_n(Z)}{\sqrt{\frac{Q}{M}(1 + Z)(c + 1)}}. \tag{49}$$

So g carries the *shifted Chebyshev weight* $1/\sqrt{(Z - Z_{\rm lo})(1 - Z)}$ on $[Z_{\rm lo}, 1]$ — the identical structure to the full window (47), only on a shifted interval — and is integrated by the *same* rule: shifted Gauss–Chebyshev, whose nodes carry the weight exactly and whose weights are the uniform π/n_θ,

$$I_n \;\approx\; \frac{4}{M}\,\frac{\pi}{n_\theta}\sum_{k=1}^{n_\theta}\frac{T_n(Z_k)}{R(Z_k)}, \qquad Z_k = m + h\cos\theta_k, \quad \theta_k = \frac{(2k-1)\pi}{2n_\theta}, \quad c_k = \frac{C + QZ_k}{M}, \tag{50}$$

m, h the midpoint and half-width of $[Z_{\rm lo}, 1]$. Equations (47) and (50) are *one formula* $I_n = \frac{4}{M}\frac{\pi}{n_\theta}\sum_k T_n(Z_k)/R(Z_k)$ on the two windows, differing only in the interval ($[-1, 1]$ vs $[Z_{\rm lo}, 1]$) and in the regular factor left in R: $R^2 = c^2 - 1$ for the full window, $R^2 = \frac{Q}{M}(1+Z)(c+1)$ for the partial. Both remove the inverse-square-root weight *analytically*, so the summand is bounded, no positivity guard is needed, and the near-edge nodes carry their correct finite contribution.

Transition zone. As $Z_{\rm lo} \to -1^+$ the smooth g acquires a square-root branch point at $Z = -1$ (from the $1 + Z$ in R) just below the lower endpoint, and the fixed-order Gauss–Chebyshev error degrades — at $Z_{\rm lo} = -0.995$, $n_\theta = 32$ gives relative error $\sim 10^{-3}$. Quadrupling the node count restores it ($\sim 10^{-12}$): the partial branch densifies for $Z_{\rm lo} < -\frac{1}{2}$, exactly as the full branch does (Eq. 47 region) on the other side of the boundary.

The algebraically equivalent alternative — forming $v_i\,T_n(Z_i)/\sqrt{(1 - Z_i^2)(c_i^2 - 1)}$ with the fused weight $v_i = \frac{\pi}{2}w_i \sin\theta_i$ of (25) carrying the $\sin\theta$ against a radicand that itself vanishes at the edge — is the same number in exact arithmetic (indeed $h\,v_i/\sqrt{(1 - Z^2)(c^2 - 1)} = \frac{\pi}{2}w_i/R(Z)$), but in floating point it is a 0/0: the radicand rounds to $\lesssim 0$ for roughly 1% of nodes (worst case $\mathcal{O}(10^{-31})$ negative), where the square root would fault or return ∞. It must then be guarded ($u > 0$ or skip), which silently forfeits those nodes. The factored form (50) needs neither the guard nor the sacrifice — $R(Z)$ is regular by construction.

5.1.3 Convergence and the transition zone

The integrand of (47), $f(Z) = T_n(Z)/\sqrt{c^2 - 1}$, has a branch point at $Z = Z_{\rm lo}$, *outside* the interval when $Z_{\rm lo} < -1$. For a function analytic on $[-1, 1]$ with nearest singularity at a real

point $Z_0 < -1$, Gauss–Chebyshev error decays geometrically at the rate of the Bernstein ellipse through Z_0,

$$\varepsilon \;\sim\; \rho^{-2n_\theta}, \qquad \rho = |Z_0| + \sqrt{Z_0^2 - 1}. \tag{51}$$

Evaluating (51) with $Z_0 = Z_{\mathrm{lo}}$ and the working order $n_\theta = 8$:

Z_{lo}	-3	-1.5	-1.2	-1.05
ρ	5.83	2.62	1.86	1.37
ρ^{-2n_θ}	$6 \cdot 10^{-13}$	$2 \cdot 10^{-7}$	$4.7 \cdot 10^{-5}$	$6.5 \cdot 10^{-3}$

The rule is excellent while the branch point stays away from the interval and degrades sharply as $Z_{\mathrm{lo}} \to -1^+$: the "transition zone". This failure mode was first seen empirically: the $n = 2$ matrix validation degraded to 2.3×10^{-3} while $n = 0$ remained at 10^{-4}. The asymmetry is instructive — for $n = 0$ the under-resolution errors are of a single sign in Z and largely average out under the subsequent q_T integration, whereas the sign-changing T_2 weight prevents the cancellation and exposes them. Per (51), quadrupling the order multiplies the exponent by four (ρ^{-8n_θ}), e.g. at $Z_{\mathrm{lo}} = -1.05$: $1.37^{-64} \approx 2 \cdot 10^{-9}$. The implementation therefore switches to $4n_\theta$ Chebyshev nodes whenever $Z_{\mathrm{lo}} > -\frac{3}{2}$, i.e. exactly where (51) predicts the base rule falls above the 10^{-7} level. The choice of threshold is a cost/accuracy compromise: a first attempt with the densification band extended to $Z_{\mathrm{lo}} > -3$ was accurate but slowed the full table build below the gain of the Z-space form itself.

5.1.4 Measured effect

In a historical full-table assembly benchmark (1518 species, 8 threads, n_q=32, n_θ=8, n_μ=8, and a 168-point grid):

assembly scheme	single 2-body build	table build	n=2 validation
$\Delta\eta$-space (arccosh + cosh + arccos)	0.51 ms	20.2 s	$2.6 \cdot 10^{-5}$
$\Delta\eta$-space, arccos $\to T_n$ recurrence	0.41 ms	20.2 s	$2.6 \cdot 10^{-5}$
Z-space, naive	0.27 ms	16.8 s	$2.3 \cdot 10^{-3}$ (fails)
Z-space, densified transition	0.33 ms	18.3 s	$1.3 \cdot 10^{-4}$

This form applies only to the assembly (and any η-independent evaluation): the functor path evaluates an arbitrary source at $\eta_p \pm \Delta\eta$ and $\phi_p \pm \alpha$, so there the rapidity and the azimuthal angle are genuine arguments and the $\Delta\eta$-space form of Sec. 2 remains the appropriate one.

6 From per-channel matrices to the full cascade

The matrix of Sec. 5 describes one observed daughter of one decay channel, for one Fourier mode. `CascadeTables.jl` turns these local maps into a full feed-down calculation. If channel c has parent $a(c)$, daughter slots $d_s(c)$, and branching ratio B_c, the cascade starts from primary spectra

$$F_{i,n} = F_{i,n}^{\mathrm{prim}} \tag{52}$$

and performs, once for every massive observed-daughter slot,

$$F_{d_s,n} \;\leftarrow\; F_{d_s,n} + B_c\, W_{\kappa(c,s),n}\, F_{a(c),n}. \tag{53}$$

The matrix is built with unit branching ratio; B_c is applied only in (53). If the same daughter occurs twice, as in $\rho \to \pi\pi$, the same matrix is deliberately applied twice. This gives the physical multiplicity without storing the matrix twice.

6.1 Channels, observed slots, and canonical keys

Three objects must be distinguished:

1. a physical channel record, containing parent and daughter identifiers and B_c;
2. an observed-daughter slot s within that record;
3. the unique kinematic key of the matrix used for that slot.

For a two-, three-, or four-body channel the key is

$$\kappa(c,s) = (N_d,\ m_a,\ m_{d_s},\ \Gamma_{\text{key}},\ m_{\text{th,key}},\ \text{sort}\{m_{d_t} : t \neq s\}), \qquad \Gamma_{\text{key}} = \begin{cases} \Gamma_a, & \Gamma_a > 10^{-3}\ \text{GeV}, \\ 0, & \text{otherwise.} \end{cases} \tag{54}$$

Here $m_{\text{th,key}}$ is zero for a sharp parent and otherwise the resolved species threshold, the lightest nonzero decay threshold in the unfiltered model. Resolving it before channel filtering keeps spectral shapes invariant and makes every filtered plan a key subset of the corresponding full table. The partner block is zero-padded to three masses in the concrete `ChannelKey`. The observed mass m_{d_s} is *not* sorted with the partners: it controls E_b, p_b, the q_T band, and the Jacobian. The unobserved partners enter symmetrically and may be sorted. Particle identifiers and B_c are intentionally absent, so charge-conjugate or equal-mass channels share matrices and a branching-ratio update requires no rebuild. Grid, quadrature orders, and requested modes are properties of the whole table rather than each key.

The function `eachchannelkey` is the single enrollment rule used both when finding matrices to build and when applying the cascade. It:

- accepts channels with two to four resolved daughters;
- requires either $m_a > \sum_i m_i + 10^{-6}\,\text{GeV}$ at the pole or, for a broad parent, $m_a + 4\Gamma_a > \sum_i m_i + 10^{-6}\,\text{GeV}$;
- invokes the operation once per massive observed slot. A photon is skipped as the observed daughter but remains a valid unobserved partner of a massive daughter;
- skips a slot whose daughter identifier equals the parent identifier, which prevents input/output aliasing in the in-place update.

One-body records are strict terminal self entries and are skipped by enrollment. Model resolution verifies their identifier, unit branching ratio, and exclusivity within the parent block. Channels with five or more daughters are rejected when constructing the runnable decay list.

6.2 Table construction and cache identity

A `CascadeTable` contains

$$\mathcal{T} = (g,\ \text{orders},\ \text{modes},\ \text{width_cut},\ \text{width_window},\ \text{spectral},\ \text{metadata},\ \mathcal{W}), \qquad \mathcal{W} : \kappa \longmapsto \{W_{\kappa,n}\}_{n\in\text{modes}}. \tag{55}$$

With the default common grid, every logical $W_{\kappa,n}$ is a 134×134 `Float32` operator stored as an offset-backed `RowBandMatrix`. Assembly accumulates each row at `Float64` precision into a

temporary dense `Float32` matrix, then packs its contiguous support span. Cascade spectra and their accumulated feed-down remain `Float64`.

Unique keys are independent build tasks. The implementation first warms the shared quadrature-node caches, then uses `Threads.@threads :greedy` over keys. Greedy scheduling matters because the recorded per-key cost ranges from roughly 0.5 ms for a sharp two-body key to 5.5 ms for a broad three-body key. All requested Fourier modes are fused within a key. The inner matrix assembly is left serial (`threaded=false`) because the outer key loop already occupies the thread pool; nested threading would oversubscribe it. Each worker holds dense matrices only for its current key, packs them immediately, and releases the dense temporaries. Packed results accumulate in a preallocated vector, and the dictionary is filled serially afterward, so the build needs no dictionary lock and never retains a second dense copy of the complete table.

Cache compatibility requires exact agreement of the format version, grid, orders, modes, width cut, width window, and spectral-shape selection. These settings are concrete table fields because they determine keys or matrix values; metadata contains provenance only. The decay-list SHA is provenance rather than a hard invalidation boundary. On a hash mismatch, matrices whose mass/width keys still match are retained, missing keys are built, and `prune!` can remove orphan keys. A branching-ratio-only change reuses every matrix. This is intentionally a lenient, incremental cache policy. In the current code, a lenient update does not refresh the stored SHA/path/raw-file metadata, and `prune!` changes only the in-memory table until the caller explicitly saves it.

6.3 Feeder-before-fed ordering

Equation (53) is an exact one-pass evaluation only if a parent is processed after all spectra that feed it have arrived and before it feeds its own daughters. Thus the required order is *feeder before fed*. For ordinary pole-open decays, daughters are lighter and this reduces to the familiar mass-descending sweep.

A finite-width tail can reopen $P \to D + x$ even when the pole mass of D is larger than that of P. A pole-mass ordering would process D too early and silently lose the downstream contribution $P \to D \to \cdots$. The implementation instead topologically sorts the enrolled feed-down graph, using the heaviest ready parent only as a tie-break. This preserves the usual order on a pole-downhill graph and handles a heavier reopened daughter correctly.

An acyclic feed graph is therefore a mathematical precondition for an exact single sweep. The implementation first collapses strongly connected components, topologically sorts the resulting acyclic component graph, and orders members of each cyclic component by mass. It warns when such a component exists; only feed-down within that component is approximate. An exact treatment would identify the strongly connected component and solve a coupled linear system of the form

$$(I - A_{\mathrm{cyc}})F_{\mathrm{cyc}} = F_{\mathrm{cyc}}^{\mathrm{prim}} + F_{\mathrm{cyc}}^{\mathrm{incoming}}, \tag{56}$$

which is not implemented.

6.4 The hot apply path and the role of dictionaries

The persistent table keeps matrices in a dictionary from `ChannelKey` to a fixed tuple of concrete `RowBandMatrix{Float32}` objects. `CascadePlan` resolves that dictionary, particle identifiers, graph order, and repeated transfers once. Execution then uses dense spectrum indices plus flat

step/group/target arrays; dictionaries remain only at the public input and output boundary. A singleton target uses the five-argument operation

$$\texttt{mul!(Fd, W, Fp, Bc, 1.0)} \quad \Longleftrightarrow \quad F_d \leftarrow B_c W F_p + F_d. \tag{57}$$

This avoids allocating a temporary WF_p. A group with several targets computes WF_p once into scratch and scatters the weighted result. The code also excludes $d = a$, so output never aliases the parent input.

The multiply is deliberately mixed precision: packed `Float32` values times a `Float64` vector accumulate into a `Float64` vector. This preserves double-precision accumulation through a long decay chain but cannot use a same-type BLAS GEMV. Because `RowBandMatrix` is a package-owned `AbstractMatrix`, it specializes Julia's standard five-argument `LinearAlgebra.mul!` interface without type piracy. The kernel walks each stored row span contiguously, converts values to `Float64`, and accumulates with a SIMD `muladd` reduction.

The grouped plan also uses a private `_cascade_axpy!` SIMD kernel to scatter a shared `Float64` scratch product to several daughters; other types fall back to `LinearAlgebra.axpy!`. All-`Float32` SGEMV was faster in a cache-hot test but introduced errors of order 3×10^{-7} per multiply. All-`Float64` DGEMV did not improve the tested dense kernel and would increase the value payload. Neither is used by the cascade.

For the bundled list there are only 44,643 slot-level operation records but a hypothetical dense two-mode sweep would perform about 1.60×10^9 matrix-element multiply–adds and stream about 6.41 GB of `Float32` values without cache reuse. The packed kernel visits only stored row spans; the production payload census is 44.98% of dense. Removing a dictionary lookup does not remove that remaining numerical work. Consequently, the dictionaries are neither the dominant RAM cost nor the inner-loop bottleneck.

Historical dense-table cascade speed (superseded). On 25 July 2026, an Apple M2 running Julia 1.12.6 with four Julia threads and one BLAS thread gave the following warmed measurements at the production grid and mode settings. The machine had only 0.06–0.08 GiB reported free, so the 2.448 GiB raw dense matrix payload was deliberately not materialized. Instead, two deterministic parent samples measured resident matrix application without risking a swap-dominated or out-of-memory run:

parents	keys	slots	table	apply median	empty median
20	730	831	100.01 MiB	16.48 ms	8.07 ms
50	1,600	1,939	219.19 MiB	28.82 ms	8.78 ms

Here "empty" constructs all 1,518 primary and output spectra but applies no decay channels. Subtracting that fixed cost and the sampled graph walk, scaling only the matrix work by the slot count, and then adding the directly measured full graph walk gives 0.473 and 0.483 s. Their agreement makes about 0.48 s a useful *warmed resident-table linear projection*, not a measured full-table time or a guaranteed lower bound: the sample pages were resident across repeated trials, whereas the full table has a much larger working set, streams up to 6.41 GB of matrix values, and is sensitive to memory pressure, paging, TLB behavior, and key reuse.

The complete 44,643-slot topology can be measured without loading any matrices. Its warmed median was 29.06 ms (minimum 27.28 ms) with 28.44 MiB allocated. Thus replacing dictionaries cannot explain the gap between the approximately 0.48 s resident projection and the repository's roughly 2 s historical full sweep; matrix access remains dominant. The earlier

in-place `mul!` change reduced historical whole-apply allocations from roughly 152 to 45 MB and improved time by about 4%, while changing results only at the 4×10^{-14} level.

Packed row-band benchmark. On 28 July 2026, 128 real production matrices at $G = 134$ were sampled from 64 evenly spaced keys. Nine alternating measurements, each applying all matrices 40 times, gave 20.45 ms for the former dense mixed kernel and 12.05 ms for standard `mul!` on `RowBandMatrix`, a factor 1.70 improvement. Packed storage was 3.906 MiB versus 8.768 MiB dense (44.56%). The maximum normalized output difference was 1.177×10^{-15} and the warmed packed call allocated zero bytes.

A separate complete-topology deterministic benchmark used all 18,299 keys, 1,518 particles, two modes, $G = 16$, two Julia threads, and one BLAS thread. The flat plan took 23.121 ms serially and 12.240 ms with modes threaded; isolated apply times were 21.213 and 10.640 ms. The keyed/plan normalized difference was 1.443×10^{-15} and serial/threaded outputs were bitwise equal. This validates orchestration and the standard-dispatch packed path, but is not a production-grid full-cache timing.

A separate build estimate sampled 24 keys from each of the six combinations of multiplicity (two, three, or four bodies) and parent treatment (sharp or broad), population-weighted the per-key work, and divided by the observed 3.57–3.67 speedup on that 144-key, four-thread sample. Two runs projected 6.53–7.65 s for all 18,299 keys. This is a sampled matrix-construction estimate (roughly 7 s, before serialization), while the 10 s recorded full build below remains the stronger end-to-end reference.

For repeated cascades with the same table and topology, `CascadePlan` maps particle identifiers to dense integer indices once, retains direct matrix references, and stores flat ordered step, group, and target arrays. Equivalent daughter multiplicities are summed into one target weight, and groups sharing one parent and matrix compute the matrix product only once. Singleton groups accumulate directly into their daughter; multi-target groups compute into one run-local scratch vector and scatter it. The table's keyed dictionary remains the construction, persistence, and filtering representation, while the plan removes key hashing and topology work from repeated numerical runs. This is a constant-factor orchestration improvement; the matrix stream remains the dominant work.

A metadata-only full-list benchmark makes the distinction precise. The `W` hash lookup contributed roughly 7–8 ms over all 44,643 slots, while a precompiled dense-index operation vector iterated in about 0.1–0.2 ms. Replacing `W` by a vector but still performing a key-to-index dictionary lookup for every slot was neutral or slightly slower: it retains hashing and adds a vector dereference. Even eliminating all measured topology and initialization overhead gives only a roughly 30–38 ms end-to-end ceiling, about 6–8% of the resident projection or 2% of the historical full sweep. The implementation therefore keeps the keyed dictionary for building, invalidation, and filtering, and compiles a separate plan for repeated applies.

The row-band experiment is now implemented: each row stores only its contiguous support span. The full-production payload estimate is 1.1012 GiB (44.98% of dense), while the representative real-matrix kernel used 44.56% of dense storage and ran 1.70 times faster.

6.5 Memory scaling

Let K be the number of unique keys, M the number of modes, G the common grid length, S the number of species, and $L_{\kappa nr}$ the stored span length of row r. Ignoring small object and dictionary headers, the payload estimates are

$$\mathcal{M}_W^{\text{dense}} = K\,M\,G^2\ \text{sizeof}(\texttt{Float32}), \tag{58}$$

$$\mathcal{M}_W^{\text{packed}} \simeq \sum_{\kappa,n}\left[4\sum_{r=1}^{G} L_{\kappa nr} + G\ \text{sizeof}(I) + (G+1)\ \text{sizeof}(O)\right], \tag{59}$$

$$\mathcal{M}_{F+F^{\text{prim}}} \simeq 2S\,M\,G\ \text{sizeof}(\texttt{Float64}). \tag{60}$$

Here I stores each row's first column and O stores sentinel value offsets; at $G = 134$ they are `UInt8` and `UInt16`. Packed storage still has a quadratic worst case, but its actual size follows the support spans rather than the dictionary count alone.

The following census was recorded for the legacy combined `decays_QM2016Plus_massorder.dat` before the canonical split format and spectral-threshold key were introduced. It is retained as a historical storage baseline, not as the current table count:

quantity	current value
species	1,518
multi-daughter channel records (plus stable self rows)	20,509 (+24)
enrolled massive observed-daughter slots	44,643
unique keys: two-/three-/four-body	12,725/5,042/532
all unique keys, of which broad	18,299, 3,227 broad
massless observed slots skipped	1,242
closed/reopened/unresolved/five-body channels	0/0/0/0

At $G = 134$ and $M = 2$, the dense baseline is

$$18{,}299 \times 2 \times 134^2 \times 4 = 2{,}628{,}614{,}752\ \text{bytes} = 2.448\ \text{GiB}. \tag{61}$$

The measured support census gives a current packed estimate of 1.1012 GiB, or 44.98% of that dense baseline. With `output=:all`, `F` and `Fprim` together contain 6,509,184 bytes (6.21 MiB) of spectrum values. The default stable output retains only 24 species, or 102,912 bytes (about 100.5 KiB), while unstable spectra remain internal for the duration of the cascade. The full apply executes 89,286 mode-specific matrix–vector products in the keyed slot sweep. Grouping in `CascadePlan` reduces this to 78,150, corresponding to 5.23 GiB of logical dense matrix traffic before cache reuse, not current physical packed traffic. At $G = 268$ the dense baseline would be 9.79 GiB; packed scaling depends on the new support spans. Adding modes grows either representation linearly.

The superseded dense cache recorded roughly 10 s to build on eight threads, 7 s to load the approximately 2.5 GB cache, and 2 s for the two-mode sweep. Packed-cache end-to-end timings remain to be measured.

6.6 Persistence when the cache approaches or exceeds RAM

The current cache uses Julia's standard `Serialization`. Saving streams the object graph to one file, but the complete packed dictionary must already be resident in memory. A raw

magic/version prefix is checked before deserialization, so a legacy dense cache is rejected without materializing its multi-gigabyte payload. A matching cache is still reconstructed in full and its row offsets are validated before use; individual keys cannot be paged selectively. The format is therefore a regenerable Julia-local cache, not a portable long-term data format.

The current writer also opens the final path directly in write mode. An interrupted save may leave a truncated file; loading catches the failure and rebuilds, but the previous good cache has already been lost. A safer writer would stream to a temporary file on the same filesystem, flush and close it, then atomically rename it over the destination.

Before a large build, Julia can expose both the predicted payload and coarse system-memory information:

```
G, M = length(grid), length(modes)
K = length(CascadeTables.neededkeys(dl)[1])
dense_upper = K * M * G^2 * sizeof(Float32)
packed_estimate = 0.4498 * dense_upper
@show dense_upper / 2.0^30         # dense upper bound
@show packed_estimate / 2.0^30     # production-census estimate
@show Sys.total_memory() / 2.0^30
@show Sys.free_memory() / 2.0^30
@show Sys.maxrss() / 2.0^30

# After a table exists:
@show Base.summarysize(tab) / 2.0^30
@show filesize("Wtables.jls") / 2.0^30
```

`Sys.free_memory()` is only a conservative instantaneous signal: the operating system may reclaim file cache, while other processes can also consume memory after the check. `Sys.maxrss()` is the process resident-set high-water mark, and `Base.summarysize` estimates Julia-owned reachable objects but not the resident pages of an mmap. A safety margin is still required.

There are three distinct persistence choices:

format	RAM behavior	speed and best use
Julia Serialization	full table reconstructed	simple and fast when it fits; current local regenerable cache
`mmap` binary	demand-paged by the OS	cold page faults, low overhead once warm; lowest-overhead out-of-core local cache
HDF5, chunked	selected chunks can be read	chunk cache, copies, and optional decompression; portable metadata and subset-oriented workflows

For a cache larger than RAM, the speed-oriented design is a small metadata index plus key-major mapped arrays for `column_starts`, `value_offsets`, and the flat `Float32` values. Keys can be built and written one at a time, so neither saving nor loading requires a second full resident copy. The compiled operation plan described above would store matrix offsets rather than dictionary-owned arrays. This design is not implemented in the current repository.

Memory mapping does not make disk I/O disappear. A full cascade touches almost every enrolled matrix, so a cold out-of-core sweep is limited by storage bandwidth and page faults;

a resident table remains faster when RAM is sufficient. Mapping is most valuable because it permits a file larger than RAM, avoids all-at-once deserialization, and lets the operating system retain the hottest pages.

HDF5 is preferable when portability, rich metadata, compression, or reading only a filtered subset matters more than minimum hot-loop overhead. A reasonable layout groups the three row-band arrays for every key and mode, with chunks aligned to whole keys or useful key batches. Splitting the arithmetic itself into smaller p_T chunks would add loop and I/O overhead while every update still consumes each stored row span. Thus *storage-level* chunking or sharding is useful; chunking the in-memory cascade multiply is not the first speed optimization. Compression must be benchmarked against the payload: smaller disk traffic can help a cold load, but decompression and copying can slow repeated full sweeps.

6.7 Filtering, validation, and regression evidence

Filtering changes the channel topology but retains all species, so a full table can apply any filtered subset without rebuilding. A lean table built from a filtered list cannot apply a later superset: every required key must already exist, otherwise the current apply raises a missing-key error. Likewise, the `primary` callback must return one grid-length vector per species and mode; excluding a primary requires `zeros(length(grid))`, not a scalar zero.

The runtime reader accepts the repository's canonical split format: particle metadata and integer-ID decay topology are parsed independently, then resolved into a strict `DecayList`. Legacy AZYHYDRO and THERMINATOR formats are offline converter inputs under `tools/`. Resolution rejects unresolved IDs, nonfinite branching fractions, invalid terminal rows, and unsupported multiplicities before table construction.

`validate(dl)` is a separate, explicit physics audit; structural model resolution has already run. It checks charge and baryon conservation, individual and summed branching ratios, duplicates, and reports strangeness/charm/bottomness changes and subthreshold rows. It does not make branching normalization a construction requirement, establish table coverage, or validate grid and quadrature choices.

The cascade regression tests currently verify:

- direct two-/three-body composition and repeated-daughter multiplicity agree with the table sweep to better than 10^{-5};
- save/load round-trips reproduce the spectra bitwise;
- adding a channel builds only its two new observed-slot keys and `prune!` removes those two orphans;
- an all-pass filter is bitwise identical to the unfiltered apply, while a two-body filter reuses the full table and removes only the excluded feed-down;
- species thresholds survive topology filtering, including a broad multi-threshold parent whose two-body-only plan reuses a full table;
- a broad parent changes the daughter feed-down relative to its sharp counterpart, and a synthetic tail-reopened parent is ordered before its heavier daughter so downstream feed-down remains nonzero.

7 Rapidity-dependent sources: the block-Toeplitz structure

This section derives the rapidity-dependent extension. The executable regression evidence in Sec. 8 applies to the currently implemented `MassAverage` functor and boost-invariant matrix path; it does not claim an implementation test of the Toeplitz construction below.

7.1 The kernel as a rapidity convolution

The boost-invariant result of Sec. 5 folded the $\pm\Delta\eta$ contributions together because the source carried no rapidity dependence. For a general source that fold is not available; instead the rapidity structure of the kernel must be exposed. Return to the mode-n decay map *before* any η-folding. The analytic azimuthal integration [Eq. (41)] gives, for a rapidity-dependent source,

$$\mathcal{F}_{b,n}(p_T,\eta_p) = \frac{B\,m_a}{4\pi p_b}\int_0^\infty \mathrm{d}q_T\, q_T \int_{-\infty}^{\infty}\mathrm{d}\Delta\eta\, \frac{\Theta\big(Q^2-D^2\big)}{\sqrt{Q^2-D^2}}\, 2T_n(Z)\, \mathcal{F}_{a,n}\big(q_T,\eta_p+\Delta\eta\big), \tag{62}$$

where the parent rapidity has been written $\eta_q = \eta_p + \Delta\eta$ and $\mathrm{d}\eta_q = \mathrm{d}\Delta\eta$ at fixed η_p; every kinematic quantity $D = M\cosh\Delta\eta - C$, $Z = D/Q$ depends on rapidity only through $\Delta\eta$. The daughter rapidity η_p therefore appears in (62) *only* as the shift of the source's argument — the statement that the decay map commutes with longitudinal boosts, $\eta_p \to \eta_p + \text{const}$ carrying η_q along with it.

Collect everything that multiplies the source into a *profile*

$$K_n(p_T,q_T,\Delta\eta) \;\equiv\; \frac{B\,m_a}{4\pi p_b}\, q_T\, \frac{\Theta\big(Q^2-D^2\big)}{\sqrt{Q^2-D^2}}\, 2T_n(Z(q_T,\Delta\eta)), \tag{63}$$

so that (62) reads

$$\mathcal{F}_{b,n}(p_T,\eta_p) = \int_0^\infty \mathrm{d}q_T \int_{-\infty}^{\infty}\mathrm{d}\Delta\eta\, K_n(p_T,q_T,\Delta\eta)\, \mathcal{F}_{a,n}(q_T,\eta_p+\Delta\eta)\,: \tag{64}$$

at each (p_T, q_T) this is a *convolution* of the source in rapidity with the fixed profile $K_n(p_T, q_T, \cdot)$. Three properties of the profile follow directly from the kernel analysis of Sec. 2.1.1 and carry the entire structure of what follows:

1. **Even:** $K_n(\,\cdot,-\Delta\eta) = K_n(\,\cdot,\Delta\eta)$, since D, Z, and $\Theta(Q^2-D^2)$ all depend on $\Delta\eta$ through $\cosh\Delta\eta$ alone. The profile is a symmetric rapidity point-spread function; it does *not* know the sign of the boost.

2. **Compact support:** K_n vanishes outside $|\Delta\eta| \in [\eta_{\rm lo}, \eta_{\rm hi}]$ (the window of Sec. 2.1.1), a set of total rapidity extent $\le 2\,\eta_{\rm hi} = 2\,\mathrm{arccosh}[(C+Q)/M]$, in practice 1–2 units.

3. **η_p-independent:** K_n depends on p_T and q_T but not on where in rapidity the daughter sits. This is boost invariance, and it is what makes (64) a genuine convolution rather than a general integral operator.

Physically $K_n(p_T, q_T, \cdot)$ is the rapidity smearing induced by the decay: a parent line at a single rapidity feeding daughter momentum p_T is spread by the profile into a band of daughter rapidities of width set by $\eta_{\rm hi}$. A boost-invariant source ($\mathcal{F}_{a,n}$ independent of η_q) is the eigenfunction with the source pulling out of the convolution, $\mathcal{F}_{b,n} = \big(\int K_n\,\mathrm{d}\Delta\eta\big)\mathcal{F}_{a,n}$, recovering Sec. 5 — there the $\int K_n\,\mathrm{d}\Delta\eta$ *is* the boost-invariant matrix element.

7.2 Discretization: the block-Toeplitz matrix

Sample both spectra on a common *uniform* rapidity grid $\eta_a = \eta_0 + a\,h_\eta$ (uniformity is essential — it is what turns the convolution into a Toeplitz operator). Index the daughter spectrum by $(k,a) \leftrightarrow (p_T^{(k)}, \eta_a)$ and the parent by $(j,b) \leftrightarrow (q_T^{(j)}, \eta_b)$. Discretize (64) in two stages. First the q_T integral and profile, by the same quadrature-plus-interpolation as Sec. 5, which produces the q_T scatter onto column index j. Second the source's rapidity argument: at a quadrature node with offset $\Delta\eta$, the value $\mathcal{F}_{a,n}(q_T^{(j)}, \eta_a + \Delta\eta)$ is reconstructed from the η grid by the uniform Lagrange stencil,

$$\mathcal{F}_{a,n}\big(q_T^{(j)}, \eta_a + \Delta\eta\big) \;\approx\; \sum_b L^\eta\big(\eta_a + \Delta\eta;\, \eta_b\big)\, \mathcal{F}_{a,n}\big(q_T^{(j)}, \eta_b\big). \tag{65}$$

The crucial observation is that on a uniform grid the cardinal weight depends only on the *offset* of the evaluation point from the node in grid units,

$$L^\eta\big(\eta_a + \Delta\eta;\, \eta_b\big) = \ell\Big(\frac{(\eta_a + \Delta\eta) - \eta_b}{h_\eta}\Big) = \ell\Big(\frac{\Delta\eta}{h_\eta} - (b-a)\Big), \tag{66}$$

a function of $b-a$ and $\Delta\eta$ but *not* of a and b separately — translation invariance of the uniform grid, inherited from translation invariance of the profile. Assembling the coefficient of $\mathcal{F}_{a,n}(q_T^{(j)}, \eta_b)$ in $\mathcal{F}_{b,n}(p_T^{(k)}, \eta_a)$ therefore yields a quantity that depends on the rapidity indices only through $d \equiv b - a$:

$$W\big[(k,a),(j,b)\big] = w\big[k,\, j,\, d\big], \qquad d = b - a, \tag{67}$$

with

$$w[k,j,d] = \int \mathrm{d}\Delta\eta\; \widetilde{K}_n^{(k,j)}(\Delta\eta)\, \ell\Big(\tfrac{\Delta\eta}{h_\eta} - d\Big), \tag{68}$$

$\widetilde{K}_n^{(k,j)}$ being the profile after the q_T scatter (the quadrature sum over the q_T nodes whose stencil hits column j). Viewing W as an $N_\eta \times N_\eta$ array of $N_{p_T} \times N_q$ blocks $B_d = w[\cdot,\cdot,d]$, (67) says W is constant along block diagonals,

$$W = \begin{pmatrix} B_0 & B_1 & B_2 & \cdots \\ B_{-1} & B_0 & B_1 & \\ B_{-2} & B_{-1} & B_0 & \\ \vdots & & & \ddots \end{pmatrix}, \tag{69}$$

a *block-Toeplitz* matrix. Storing W densely repeats each block up to N_η times; storing the tensor $w[k,j,d]$ once is exact and lossless — it is the discrete profile, the point-spread function of Sec. 7.1 sampled at integer rapidity offsets.

Two further exact properties shrink the tensor, both inherited from the profile:

1. **Banded:** compact support of K_n (item 2 above) plus the finite reach of the η stencil limits the nonzero offsets to $|d| \le L$ with $L = \lceil \eta_{\mathrm{hi}}^{\max}/h_\eta \rceil + \frac{P}{2}$ (window half-width plus stencil halo), $\eta_{\mathrm{hi}}^{\max} = \max_{q_T} \operatorname{arccosh}[(C+Q)/M]$, in practice 1–2 units; so $B_d = 0$ for $|d| > L$ and only $2L+1$ offsets exist.

2. **Even in d:** evenness of the profile (item 1) gives $w[k,j,-d] = w[k,j,d]$, so only $d \ge 0$ need be stored. (The *source* need not be symmetric in rapidity; the *profile* is.)

For $N_{p_T} = 39$, $N_q = 121$, $N_\eta = 61$, $h_\eta = 0.05$, $L \approx 40$: dense W is $(39{\cdot}61){\times}(121{\cdot}61) \approx 1.76{\times}10^7$ entries ($\sim$140 MB), the tensor is $39 \times 121 \times 41 \approx 1.9 \times 10^5$ entries ($\sim$1.5 MB) — a factor $\sim 10^2$, at zero accuracy cost, because it encodes exact symmetries (translation invariance and parity of the profile) rather than a numerical truncation.

7.3 Assembly

The assembly generalizes the scatter of Sec. 5 by one stencil dimension. Fix a Fourier mode n and run the same $(q_T, \Delta\eta)$ quadrature loops as the kernel (in $\Delta\eta$ space — see the remark below); a node at $(q_T, \Delta\eta)$ with total quadrature factor

$$\mathrm{fac} = \mathrm{pref} \cdot v_i^q h_q \, q_T \, h_\eta \, \frac{v_j^\theta}{\sqrt{Q^2 - D^2}} \, 2T_n(Z) \tag{70}$$

(the 2 is the two azimuthal roots only — the $\pm\Delta\eta$ fold is *not* performed here, each sign now lands on different η columns) deposits into the tensor via the product of the q_T stencil and the η-offset stencil:

$$w[k,\, j_0 + m - 1,\, d_0 + m' - 1] \mathrel{+}= \mathrm{fac}\; L_m^q(q_T)\; L_{m'}^\eta\big(\Delta\eta/h_\eta\big), \qquad m, m' = 1, \ldots, P, \tag{71}$$

where d_0 is the stencil start of the offset $\Delta\eta/h_\eta$ on the integer offset axis. Since the kernel factor is even in $\Delta\eta$, only $\Delta\eta \geq 0$ need be traversed and the result stored for $d \geq 0$; the negative offsets are recovered at application time. No loop over η_p exists anywhere: one sweep per (k, j) column of nodes builds the profile valid for *every* daughter rapidity, so the assembly cost equals that of the boost-invariant matrix times the P-fold extra stencil work — *not* times N_η. This is the payoff of the Toeplitz structure on the assembly side.

Two remarks. First, the Z-space form of Sec. 5.1 does *not* apply here: the η-offset stencil needs the rapidity $\Delta\eta$ itself, so the assembly stays in $\Delta\eta$ space (or would have to pay an arccosh per node to recover $\Delta\eta$ from c — exactly the function the Z form eliminates). Second, the offset variable is interpolated on the *uniform* η grid, so the uniform-grid Lagrange weights apply verbatim; the interpolation error in the η direction is the same $\mathcal{O}(h_\eta^P)$ as in q_T.

7.4 Application

Using the evenness $w[k, j, -d] = w[k, j, d]$, the apply is a folded discrete convolution over the stored $d \geq 0$ half:

$$\mathcal{F}_b\big(p_T^{(k)}, \eta_a\big) = \sum_j \Big[\, w[k,j,0]\, \mathcal{F}_a\big(q_T^{(j)}, \eta_a\big) + \sum_{d=1}^{L} w[k,j,d] \, \big(\mathcal{F}_a\big(q_T^{(j)}, \eta_{a+d}\big) + \mathcal{F}_a\big(q_T^{(j)}, \eta_{a-d}\big)\big)\Big]. \tag{72}$$

Organizing the computation per offset makes each term a dense $(N_{p_T} \times N_q)$ matrix acting on an η-shifted copy of the source — $L + 1$ small GEMMs over the $(q_T{\times}\eta)$ source array:

$$F_b \;=\; \sum_{d=0}^{L} B_d \left[S^{+d} F_a + (1 - \delta_{d0})\, S^{-d} F_a\right], \qquad B_d = w[\,\cdot, \cdot, d\,], \tag{73}$$

with $S^{\pm d}$ the shift along η. Cost: $\mathcal{O}(N_{p_T} N_q N_\eta L)$ multiply–adds ($\approx 2 \times 10^7 \sim$ milliseconds at N_{p_T}=39, N_q=121, N_η=61, $L{\approx}40$), organized as BLAS-3 rather than the BLAS-2 of a dense block

matrix. For large N_η the Toeplitz structure admits FFT application along η, $\mathcal{O}(N_{p_T} N_q N_\eta \log N_\eta)$ — worthwhile only once $L \gg \log N_\eta$, i.e. for very fine rapidity grids; at the sizes above the direct fold wins.

For azimuthally asymmetric, rapidity-dependent sources the procedure is per mode as before: one tensor w_n per Fourier mode, assembled with $2T_n(Z)$, applied to the mode functions $\mathcal{F}_{a,n}(q_T, \eta_q)$.

7.5 Edges and grid layout

Translation invariance fails only at the ends of the rapidity grid: daughter points within $\Delta\eta_{\max}$ of the boundary require parent values beyond it. Since physical sources vanish rapidly at large $|\eta|$ (e.g. $e^{-m_T \cosh\eta/T}$), the consistent convention is zero-padding — the same "source vanishes off the grid" convention used in q_T — with the η grid extended by $\Delta\eta_{\max} = \max_{q_T} \operatorname{arccosh}[(C+Q)/M]$ (evaluated at the band edge; typically 1–2 units) beyond the rapidity window where results are needed. Concretely: to output $|\eta_p| \le \eta^{\text{obs}}$, the parent grid must span $|\eta_q| \lesssim \eta^{\text{obs}} + \Delta\eta_{\max}$, and the truncation error inherits the source's decay at the padding edge — for thermal-like sources this is $e^{-m_{aT}(\cosh(\eta^{\text{obs}}+\Delta\eta_{\max})-1)/T}$, negligible already for modest padding.

The memory layout follows the apply: storing w as (k,j)-major slices per offset d (i.e. $L{+}1$ dense $N_{p_T} \times N_q$ matrices B_d) makes each term of (72) a contiguous GEMM; `Float32` storage is appropriate for the same reason as in Sec. 5 (rounding 6×10^{-8}, far below the quadrature setting), halving the $\sim$ 1–3 MB per decay and mode.

8 Implementation validation and domain of applicability

The results in this section concern the implemented boost-invariant path. That implementation has two distinct numerical layers. A `MassAverage` evaluates the continuous decay integral directly, while `DecayMatrix` or `decaymatrices` additionally interpolates the parent spectrum from a finite q_T grid. A useful validation therefore checks the integral before checking the matrix: otherwise interpolation and grid-truncation errors can be mistaken for errors in the decay kernel.

8.1 Validated operating domain

For every sharp two-body kernel contained in a two-, three-, or four-body average, the numerically supported domain is the *strictly open* physical region

$$m_a > m_b + \sum_{i \neq b} m_i, \qquad m_b > 0, \qquad m_i \ge 0. \tag{74}$$

The strict inequality matters. At equality the rest-frame daughter momentum p_b vanishes. The two-body prefactor contains $1/p_b$, and the normalization of a multi-body phase-space average also vanishes. The limiting decay distribution can be discussed analytically, but it is not represented by the fixed-order formulas implemented here. Branching ratios should be finite and non-negative, and all quadrature orders $n_q, n_\theta, n_\mu, n_\sigma, n_{m_a}$ must be positive integers. The observed daughter is required to have nonzero mass; a photon may occur among the unobserved partners, but the present (p_T, η) kernel does not return a massless observed spectrum.

For a finite-width parent the tested Breit–Wigner setup additionally obeys

$$m_R > 0, \qquad \Gamma > 0, \qquad 0 < m_{\min} < m_{\max}, \tag{75}$$

normally with $m_{\min} < m_R < m_{\max}$. The spectral average is normalized on the full window $[m_{\min}, m_{\max}]$, whereas decay-kernel nodes are placed only in the open portion of that window. Consequently a channel that is closed at the pole but open in the high-mass tail retains the correct open-spectral-weight suppression.

The bundled cascade applies two additional guards before constructing a matrix: a sharp channel is enrolled only if $m_a > \sum_i m_i + 10^{-6}\,\mathrm{GeV}$, and a broad parent's lower support is clipped to

$$m_{\min} = \max(m_R - 4\Gamma,\ 0.05\,\mathrm{GeV}). \tag{76}$$

Thus the exact-threshold and non-positive-support cases discussed below are excluded by the standard cascade path. They remain relevant to users calling the exported decay constructors directly.

8.2 Independent numerical checks

For the invariant number functional

$$\mathcal{N}[F] = \int_0^\infty q_T\,\mathrm{d}q_T \int_{-\infty}^\infty \mathrm{d}\eta \int_0^{2\pi} \mathrm{d}\phi\, F(q_T, \eta, \phi), \tag{77}$$

an isotropic channel must give $\mathcal{N}[F_b] = B\mathcal{N}[F_a]$. This is a particularly strong test because it integrates the returned spectrum over the entire daughter phase space. The current `runtests.jl` suite, evaluated with Julia 1.12.6, gives:

check	observable	measured result
$\rho \to \pi\pi$, sharp two-body	$\mathcal{N}_b/(B\mathcal{N}_a)$	0.999999
$\omega \to 3\pi$, three-body	$\mathcal{N}_b/(B\mathcal{N}_a)$	1.000069
$1.3\,\mathrm{GeV} \to 4\pi$, four-body	$\mathcal{N}_b/(B\mathcal{N}_a)$	1.000001
broad $\rho \to \pi\pi$	$\mathcal{N}_b/(B\mathcal{N}_a)$	1.00009
direct azimuthal result vs. $n = 0, \pm 2$ reconstruction	relative difference	at most 10^{-6}
functor orders doubled	two- and three-body relative difference	below 5×10^{-7}

These tests use an independent adaptive outer integration of a static Boltzmann source. Four-body permutations of the three unobserved cluster masses also agree within the regression tolerance, checking the nested Dalitz-area integral rather than merely its normalization.

The sampled matrix is compared separately with the continuous functor. On $q_T \in [0, 4.2]$ with 121 parent-grid points and fourth-order local Lagrange interpolation, the maximum relative differences are

channel and mode	$\rho,\ n = 0$	$\rho,\ n = 2$	$\omega,\ n = 0$
maximum relative difference	2.6×10^{-5}	2.7×10^{-4}	3.1×10^{-5}

This comparison includes interpolation and the finite parent-grid cutoff; it is therefore expected to be less precise than the direct quadrature comparison. Matrix assembly for the tested modes is deterministic: `threaded=true` and `threaded=false` produce bitwise-identical matrices. Negative Fourier indices reuse the even Chebyshev kernel ($W_{-n} = W_n$), and the regression suite also checks non-default integer mode types such as `Int8` and `Int32`. These results exercise selected thermal sources and kinematic points; they are regression evidence, not a proof for every possible input function or grid.

8.3 Production-order sampled convergence

The preceding continuous checks use the conservative functor defaults, while the cascade builds at $(n_q, n_\theta, n_\mu, n_\sigma, n_{m_a}) = (32, 8, 8, 8, 8)$. To measure that distinction directly, the bundled list was divided into the six two-/three-/four-body, sharp/broad strata. Six reproducibly selected keys from each stratum were evaluated on the production 134-point $q_T \in [0, 4]$ grid and ten daughter momenta from 0.1 to 2.0 GeV. The source was a $T = 0.145$ GeV transverse Boltzmann spectrum with parent $v_2 = 0.05$. Production matrices were compared with $(128, 32, 32, 32, 32)$ matrices of the same grid and interpolation order.

For mode zero, let ϵ_0 be the maximum absolute spectrum difference normalized by the reference peak. For mode two, a relative error can become arbitrarily large when the true response crosses zero, so the physical metric is instead the maximum absolute error in the daughter $v_2 = F_2/F_0$:

key stratum	median ϵ_0	max. ϵ_0	median $\lvert\Delta v_2\rvert$	max. $\lvert\Delta v_2\rvert$
two-body, sharp	1.05×10^{-4}	1.81×10^{-4}	8.35×10^{-6}	1.39×10^{-5}
two-body, broad	5.81×10^{-4}	8.66×10^{-3}	8.28×10^{-4}	2.77×10^{-3}
three-body, sharp	2.43×10^{-3}	2.39×10^{-2}	1.63×10^{-3}	7.27×10^{-3}
three-body, broad	4.02×10^{-3}	1.37×10^{-2}	1.93×10^{-3}	4.66×10^{-3}
four-body, sharp	2.62×10^{-2}	9.50×10^{-2}	9.14×10^{-3}	1.43×10^{-2}
four-body, broad	1.25×10^{-2}	2.22×10^{-2}	4.46×10^{-3}	7.11×10^{-3}

These are small per-stratum samples, not a population-weighted full-cascade error bar. Nevertheless, checking the worst sampled key in each stratum against $(256, 64, 64, 64, 64)$ left only $2.4\,10^{-6}$–$8.8\,10^{-6}$ residual mode-zero error and at most $5.2\,10^{-4}$ residual absolute v_2 error, so the observed production-order differences are not reference noise.

Rounding each matrix to `Float32` is not the limitation: over all 72 sampled key–mode applications its median effect was $5.9\,10^{-9}$ and its maximum was $2.5\,10^{-8}$. The convergence study instead isolates $n_\mu = 8$ as the main three-/four-body limitation and $n_{m_a} = 8$ as the additional broad-parent limitation; increasing n_q, n_θ, or n_σ alone had little effect in the difficult sampled cases.

A selective candidate $(n_q, n_\theta, n_\mu, n_\sigma, n_{m_a}) = (32, 8, 16, 8, 16)$ was checked on the worst sampled mode-zero key of every stratum. The largest mode-zero error fell from $9.50\,10^{-2}$ to $8.51\,10^{-4}$, and the largest absolute daughter-v_2 error fell from $1.39\,10^{-2}$ to $3.38\,10^{-3}$. This changes only one-time construction work; matrix size and cascade-apply speed are unchanged. It is therefore the first accuracy setting to try when multi-body anisotropy matters, followed by an order-doubling check for the specific observables.

There is still no single independently measured "full-cascade percentage": the answer depends on branching ratios, the primary spectra, cancellations, and how errors propagate through successive generations. The current cascade-vs-direct regression uses the same low-order kernels on both sides, so it validates branching ratios, multiplicity, ordering, persistence, and `Float32` wiring rather than production quadrature accuracy. A high-order end-to-end cascade comparison remains desirable.

8.4 Finite-width threshold stress test

The difficult finite-width case is a narrow open sliver: the pole lies below threshold and only the upper end of the spectral window can decay. Sampling the whole window with ordinary nodes and discarding closed nodes can miss this sliver completely. The implementation instead splits

the parent-mass quadrature at threshold and places all decay nodes in the open interval. For the deliberately severe test

$$[m_{\min}, m_{\max}] = [0.6, 1.4]\,\mathrm{GeV}, \qquad m_{\text{threshold}} = 1.3\,\mathrm{GeV}, \tag{78}$$

`nma=8` now gives a nonzero matrix. Relative to an `nma=96` reference, the integrated matrix deviation decreases from approximately 0.15 at `nma=8` to 0.04 at `nma=32`. Therefore threshold splitting fixes the silent-zero failure, but does not make a very small open fraction cheap to resolve. For custom decay lists with such channels, `nma` should be increased until the result is stable; the default is intended for ordinary, well-open windows. The bundled decay list currently enrolls no tail-reopened channels; the stress case is retained to protect custom lists and future data updates.

8.5 Known boundary behavior and recommended safeguards

Two classes of input are now rejected by the library itself, at construction. The sharp constructors test openness strictly, so exact equality in (74) raises rather than producing an infinite two-body coefficient or a `NaN` three- or four-body normalization; and `BreitWigner` requires finite arguments with $m_R > 0$, $\Gamma > 0$ and $0 \le m_{\min} < m_{\max}$, as in (75), while the five quadrature orders are required to be at least one. These were not cosmetic omissions: a non-positive width, a non-positive pole mass or a reversed interval each yielded an *empty* parent-mass average, that is a channel whose matrix is identically zero, which feeds nothing downstream and reports nothing. A negative width additionally inverted the support interval on its own. The keyword form still clamps a negative $m_{\min}$ to zero, which is a legitimate case: fourteen states in the QM2016 list are broad enough for the default $m_R - k\Gamma$ to fall below zero.

The following remain outside the validated domain and must be handled by the calling application:

1. The arrays inside `MassAverage` — its node lists, and the quadrature rules reached through `kern` — are treated as immutable by the library and are shared between every channel built from the same `Rules`. Sharing makes them safe for concurrent reads, and they are built eagerly before any task starts, so no reader can observe a partial one; but the objects are exported, and user code must not mutate them. Mutating one would corrupt every channel that shares it, not only the channel at hand.

2. For a broad channel whose open part is only a small fraction of its support, convergence in `nma` must be checked separately from convergence in $n_q, n_\theta, n_\mu, n_\sigma$. Increasing the kernel orders cannot repair an under-resolved parent-mass average.

3. A matrix represents the parent source only on its supplied q_T grid. The interpolation convention is zero outside that grid, so its upper endpoint must be placed where the omitted parent tail is negligible. This is a matrix domain choice, not an error of the continuous decay functor.

Of the checks a robust production setup needs, the library now performs those it can state locally: masses against the strict threshold, widths and support bounds through `BreitWigner`, the quadrature orders, and the rejection of any non-finite coefficient arising from a near-threshold node of a broad parent. What it cannot check is what only the caller knows — that the branching ratios are the intended ones, that the q_T grid extends past the parent tail, and that `nma` has been

converged separately for a narrowly reopened channel. Number conservation and matrix–functor agreement should be kept as regression tests regardless. None of this touches the fast path: once valid matrices have been built, feed-down remains a sequence of in-place matrix–vector products.

9 Summary

case	object	size / cost
η-independent, per mode n	logical dense operator W_n with row-band cascade storage	$\sim$38 KiB for 39×121 `Float64` ($\sim$70 KiB for 134^2 `Float32`); apply $\sim \mu$s
full cascade, current defaults	18,299 keys, two modes on 134^2	2.4481 GiB dense baseline; 1.1012 GiB estimated packed payload (44.98%); production full-apply timing pending
η-dependent, per mode n	tensor $w[k, j, d]$, $\lvert d\rvert \leq L$, $d \geq 0$	$\sim$0.8 MiB for $39 \times 121 \times 41$ `Float32`; apply $\sim$ms